

Harmonic Shift Singularities and Exact Moment Identities

Spectral derivatives, polygamma moment spaces, harmonic powers,
and the finite constants of colliding periodic singularities

Research continuation for the ProveIt project
Prepared for Vladimir Reshetnikov

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Abstract

We continue the exact-identity programme of the ProveIt manuscript and its incoming reports at a fixed repository revision. Four related problems admit rigorous extensions. First, a Gamma-normalized harmonic interpolation has an explicit singular germ at every nonpositive shift. This determines its regular boundary constants in Arakawa–Kaneko functions, its exact shift Taylor radii, and a functional rank jump: at spectral order $-m$ the depth family has rank $m + 1$, whereas any positive spectral derivative restores infinite rank. Normalized local primitives connect the same germs to the Lerch transcendent and polylogarithms. Second, the mixed Bell rows generated by a Dougall identity have exact rank $\lfloor w/2 \rfloor$ in weight w ; their ordinary summation image has rank three for $w \geq 6$. Explicit rational elimination at weights six and eight yields a quartic polygamma identity and an elementary generalized-harmonic tail-square evaluation. Third, a triangular exponent generator proves that, for fixed p , all centered values $F_p(-2r, 1/2)$ lie in one finite rational span of convergent Euler sums of weight at most $p - 1$, independent of r . We obtain complete power generators at $0, -2, -4$, all-moment Bernoulli formulas for $p = 1, 2$, and exact exponent poles. A sufficient parity class and finite ordinary-zeta formulas extend the calculation to odd generalized harmonic orders. Fourth, a finite principal-part formula for colliding periodic polygamma factors leaves a jointly holomorphic remainder. It gives exact constants for nested collision paths, arbitrary argument-derivative orders, and several fully evaluated examples involving Stieltjes constants.

The proofs specify branches, continuation order, cutoff coordinates, and the meaning of every rank assertion. Source specializations are identified explicitly. Exact symbolic checks and independent numerical diagnostics accompany the article; no numerical fit is used as a proof. The current Gaussian S_6 and revised S_8 conjectures remain open. We conclude with precise integration notes and further research targets.

Keywords. Polylogarithm; Hurwitz zeta; generalized Stieltjes constant; polygamma moment; harmonic power; analytic continuation; Hadamard finite part; multiple zeta value.

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1 Scope, conventions, and the precise advances

The aim is to turn several questions already present in the ProveIt research stream into exact theorems. The most direct antecedents are the mixed spectral report, the independent-order Hurwitz report, and the raw harmonic-power section of the exact resonant report [2, 4, 5]. These sources already contain substantial analytic continuation, Gamma normalization, and finite-part machinery. We use it with explicit attribution, supply self-contained proofs of the identities needed here, and resolve specific questions about singularities, spans, and collision constants.

The literature background is classical: Gamma and polygamma recurrences, Hurwitz-zeta continuation, Dougall’s summation, and polylogarithmic Mellin transforms [11, 10, 13, 14]. The Arakawa–Kaneko functions are established objects [8, 9]. Our claims of an advance are relative to the inspected ProveIt material. We do not claim that every special-value consequence is absent from the wider literature.

1.1 What is established

Problem	Antecedent	Result of this continuation
Shift singularities	A depth-dependent continuation half-plane; the broader relative ordered Hurwitz character and its transport.	Exact local amplitudes, boundary constants, sharp Taylor radii, normalized primitives, and the finite-to-infinite functional rank transition; Section 2.
Polygamma moments	A two-parameter Dougall identity and the request to separate moments at weights six and eight.	The full mixed-Bell rank in every weight, exact elimination matrices, the three-dimensional summation image, and its full kernel; Section 3.
Centered harmonic powers	Regularity of $F_p(-2r, 1/2)$ and a multiple-zeta expression obtained by finite word calculations.	A triangular generator for every r and all p , and one finite Euler-sum space independent of r at fixed p ; Section 4.
Geometric collisions	Fixed-rate triple digamma formulas and questions about nested rates and higher argument derivatives.	An exact all-factor polygamma correction with jointly holomorphic remainder, oriented harmonic hierarchy laws, and explicit derivative examples; Section 5.

The harmonic generator answers the all-power minus-two target and gives every further even moment by finitely many triangular steps. It is not a single closed two-variable expression in both the power and the moment order. A companion result proves regularity for polynomials in odd generalized harmonic orders and evaluates each single odd-order family in a fixed finite space of even zeta values. The collision theorem treats Stieltjes index zero with arbitrary argument derivatives. Positive Stieltjes indices, which introduce logarithmic local singularities, remain a separate extension. The Bell rank concerns the stated formal polynomial rows; it does not decide arithmetic independence of special values. These distinctions delimit what has actually been proved.

1.2 Functions and derivative conventions

We use the ordinary Gamma function, $\psi = \Gamma'/\Gamma$, and $\psi^{(q)} = \partial_a^q \psi$. The generalized harmonic numbers are

$$H_n^{(q)}(a) = \sum_{j=0}^{n-1} (a+j)^{-q}, \quad H_n^{(q)} = H_n^{(q)}(1), \quad H_n = H_n^{(1)}.$$

For an integer $q \geq 1$ they satisfy

$$H_n^{(q)}(a) = \frac{(-1)^{q-1}}{(q-1)!} \{ \psi^{(q-1)}(a+n) - \psi^{(q-1)}(a) \}. \quad (1.1)$$

The same formulas extend in a away from their poles, with branches specified when nonintegral powers occur. Relation (1.1) follows by telescoping the polygamma recurrence [11].

For $\Re s > 1$ and $a > 0$, $\zeta(s, a) = \sum_{n \geq 0} (n+a)^{-s}$, and $\zeta(s) = \zeta(s, 1)$. Generalized Stieltjes constants have the fixed convention

$$\zeta(1+v, a) = \frac{1}{v} + \sum_{j \geq 0} \frac{(-1)^j}{j!} \gamma_j(a) v^j, \quad \gamma_0(a) = -\psi(a), \quad \gamma_j = \gamma_j(1). \quad (1.2)$$

Thus $\gamma_j^{(q)}(a)$ always denotes an argument derivative, whereas $\partial_s^k \zeta(s, a)$ is a spectral derivative. The index j in γ_j is discrete and is never differentiated. Similarly, a formula known at integer spectral labels $s = -m$ cannot be differentiated with respect to m to infer a spectral derivative.

Our multiple-polylogarithm convention is

$$\text{Li}_{k_1, \dots, k_d}(z) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z^{n_1}}{n_1^{k_1} \dots n_d^{k_d}}.$$

For admissible indices, its value at $z = 1$ is $\zeta(k_1, \dots, k_d)$; weight means $k_1 + \dots + k_d$. The one-index case is the ordinary polylogarithm [14]. The Lerch transcendent is initially

$$\Phi(z, s, a) = \sum_{n \geq 0} \frac{z^n}{(n+a)^s}, \quad |z| < 1,$$

with analytic continuation where required [15]. In particular $z\Phi(z, s, 1) = \text{Li}_s(z)$.

1.3 Three different meanings of a displayed infinite expression

Most sums in Section 3, including the tail-square identity, converge absolutely in the ordinary sense. They need no regularization. In contrast,

$$F_p(-2r, 1/2)$$

means the value of the meromorphic continuation of $\sum_{n \geq 0} H_n^p(n+1/2)^{-s}$ at the regular point $s = -2r$. For $p \geq 1$ this is not the sum of an ordinarily convergent numerical series after direct substitution of $s = -2r$.

For a function with a local Laurent-log expansion, CT_z extracts the coefficient of $z^0(\log z)^0$. A Hadamard finite-part integral is the constant term in its cutoff expansion with the cutoff coordinate held fixed. At distinct interior seams we use independent ambient coordinates $x - x_i$ and take their joint constant term before a collision parameter tends to zero. The subsequent CT_ϵ in Section 5 is a second operation, on the resulting function of the geometric shifts. Changing either coordinate can change this constant. The exact correction formulas record that dependence.

Branches are also part of the statements. A principal shift continuation, a continuation around a negative integer, and a real collision path need not have the same analytic germ. All local powers are written using an explicitly chosen logarithm. Fixed-endpoint primitives are normalized to vanish at their stated base point, so an integration constant cannot absorb a resonance discrepancy.

1.4 Proof strategy and verification

Each family is handled by first retaining a full analytic parameter family, then making the desired specialization. Finite transport isolates a shift singularity; a Gamma quotient generates the Bell moments; an exponent parameter linearizes raw harmonic powers; and partial fractions isolate the singular collision pieces. Normal convergence or a holomorphic remainder is proved before extracting coefficients. This order of operations is essential at resonances.

Exact symbolic checks replay coefficient identities and rational linear algebra. Independent numerical evaluations start from different representations and diagnose signs, constants, branches, and omitted terms. They supplement the proofs. The numerical tails and quadratures are not interval certificates, and their small residuals are not used to justify an infinite identity. Section 6 records the actual checks and the boundaries of the source audit.

2 Exact shift singularities of the entire harmonic difference

The Newton expansion in the mixed-spectral report [2] proves holomorphy of its depth- r entire difference on $\Re a > -r$, and asks for the actual singularity at $a = -r$. We determine every negative-integral shift germ, the exact Taylor radius, and the finite regular term at the first singularity. The proof uses a finite shift identity and a cancellation at $a = 1$. Finite transport is a classical special-function method. The independent-orders report [4] supplies a broader entire ordered interpolation of which this remainder is an exact specialization, identified below. The contribution here is the explicit singular amplitudes and their boundary, Taylor-radius, and functional-rank consequences; we do not assert priority for the underlying transport method.

2.1 Definitions and the finite shift identity

For $\Re a > 0$, define, initially for $\Re s > 1 + \Re u$,

$$F_a(s, u) = \sum_{n \geq 0} \frac{(a+u)_n}{(a)_n(n+a)^s}, \quad K_a(u) = \frac{\Gamma(a)\Gamma(1+u)}{\Gamma(a+u)}, \quad (2.1)$$

$$D(s, u; a) = F_a(s, u) - K_a(u)F_1(s, u), \quad D_r(s; a) = [u^r]D(s, u; a). \quad (2.2)$$

Here $(x)_n = x(x+1)\cdots(x+n-1)$ is a rising factorial. Every statement in u can be read on a fixed disk $|u| < \rho < 1$. The sources [3, 2] prove that D is entire in s , and that its coefficients are jointly holomorphic for $s \in \mathbb{C}$, $\Re a > -r$. In particular,

$$D(s, u; 1) = 0, \quad D_0(s; a) = \zeta(s, a) - \zeta(s). \quad (2.3)$$

The pole of the last difference at $s = 1$ is removable.

Put

$$P_n(a, u) = \frac{(a+u)_n}{(a)_n} = \sum_{j=0}^n e_j(n; a)u^j, \quad e_j(n; a) = \sum_{0 \leq \nu_1 < \cdots < \nu_j < n} \prod_{i=1}^j \frac{1}{a + \nu_i}. \quad (2.4)$$

The empty product is one and $e_j(n; a) = 0$ for $j > n$.

Exact identification with the ordered endpoint interpolation. Let $\mathcal{H}_{b,a}(w)$ denote the relative character in [4], normalized by the deconcatenation identity $Z_a(w) = \sum_{w=vw'} Z_b(v)\mathcal{H}_{b,a}(w')$ for the strict ordered Hurwitz sums. Its Gamma formula for words of ones is $\mathcal{H}_{1,a}(\{1\}^j) = [u^j]K_a(u)$. Applying the defining recursion to $(s, \{1\}^r)$ therefore gives the exact identification

$$D_r(s; a) = \mathcal{H}_{1,a}(s, \{1\}^r). \quad (2.5)$$

Indeed $Z_a(s, \{1\}^r) = \sum_{n \geq 0} e_r(n; a)(n + a)^{-s}$, and the other deconcatenation terms are exactly the coefficients of $K_a(u)\bar{F}_1(s, u)$ in (2.2). This first proves the identity in a convergence domain, then everywhere by the entire spectral continuations. Proposition 2.1 is accordingly also a specialization of that source's endpoint transport.

Proposition 2.1 (Exact finite transport). *For every integer $L \geq 1$,*

$$D(s, u; a) = \sum_{n=0}^{L-1} P_n(a, u)(a + n)^{-s} + P_L(a, u)D(s, u; a + L). \quad (2.6)$$

Consequently

$$D(s, u; a) = a^{-s} + \frac{a + u}{a} D(s, u; a + 1), \quad (2.7)$$

$$D_r(s; a) = \mathbf{1}_{r=0} a^{-s} + D_r(s; a + 1) + \mathbf{1}_{r \geq 1} \frac{D_{r-1}(s; a + 1)}{a}, \quad (2.8)$$

$$D_r(s; a) = \sum_{n=r}^{L-1} e_r(n; a)(a + n)^{-s} + \sum_{j=0}^{\min(r, L)} e_j(L; a) D_{r-j}(s; a + L). \quad (2.9)$$

The finite sum in the last formula is empty when $L \leq r$. These identities hold wherever the powers and continuations on both sides use the same branches.

Proof. Separating the $n = 0$ summand gives $F_a = a^{-s} + (a + u)F_{a+1}/a$. The Gamma functional equation gives $K_a = (a + u)K_{a+1}/a$. Subtraction proves (2.7) in the initial common half-plane. Both sides are entire in s , so that identity continues to every s . Iterating it gives (2.6); coefficient extraction gives the other formulas. All operations are finite after the initial identity, so they create no interchange of a divergent sum and a derivative. $\square$

2.2 Every negative-integral shift germ

For $N \geq 0$, put $a = -N + z$ and define

$$C_N(z, u) = P_N(-N + z, u) = \prod_{j=1}^N \left(1 + \frac{u}{z - j}\right), \quad c_{N,r}(z) = [u^r]C_N(z, u). \quad (2.10)$$

Thus $C_0 = 1$ and $c_{N,r} = 0$ if $r > N$. Near $z = 0$ choose a branch of $\log z$. For the other powers in (2.6), choose logarithms of $z - h$, $1 \leq h \leq N$, holomorphic on a small disk around zero. Their values at zero may be $\log h + i\pi$ or $\log h - i\pi$ for the two lateral continuations.

Theorem 2.2 (Exact local decomposition). *On these specified branches,*

$$D(s, u; -N + z) = C_N(z, u)z^{-s} + H_N(s, u; z), \quad (2.11)$$

where H_N is jointly holomorphic in s , u , and z near every point $(s_0, 0, 0)$. An explicit formula is

$$\begin{aligned} H_N(s, u; z) &= \sum_{n=0}^{N-1} P_n(-N + z, u)(z - N + n)^{-s} \\ &\quad + \left(1 + \frac{u}{z}\right) C_N(z, u) D(s, u; 1 + z). \end{aligned} \quad (2.12)$$

The apparent pole in the second line is removable. For every $r, k \geq 0$,

$$\partial_s^k D_r(s; -N + z) = c_{N,r}(z)z^{-s}(-\log z)^k + \partial_s^k H_{N,r}(s; z), \quad H_{N,r} = [u^r]H_N. \quad (2.13)$$

For $r \leq N$ the leading coefficient is nonzero:

$$c_{N,r}(0) = (-1)^r e_r \left(1, \frac{1}{2}, \dots, \frac{1}{N} \right). \quad (2.14)$$

For $r > N$ the entire singular amplitude vanishes identically.

Proof. Use (2.6) with $L = N + 1$. Its $n = N$ term is exactly $C_N(z, u)z^{-s}$, and the other terms give (2.12) because $P_{N+1}(-N + z, u) = (1 + u/z)C_N(z, u)$. Every factor of P_n , $n \leq N$, is holomorphic in z near zero. The chosen logarithms make all powers with $n < N$ holomorphic there as well. By (2.3), $D(s, u; 1 + z) = z \tilde{D}(s, u; z)$ with $\tilde{D}$ jointly holomorphic. This cancels the sole $1/z$ factor in the last line. This also proves local normality on compact spectral sets, so all the stated derivatives can be taken. Finally, $C_N(0, u) = \prod_{j=1}^N (1 - u/j)$, which proves (2.14); its elementary symmetric coefficients are strictly positive before the displayed sign. $\square$

Corollary 2.3 (Principal continuation). *For each $r \geq 0$, $D_r(s; a)$ has a jointly holomorphic principal continuation to*

$$(s, a) \in \mathbb{C} \times (\mathbb{C} \setminus (-\infty, -r]). \quad (2.15)$$

It is holomorphic across $a = 0, -1, \dots, -(r-1)$. On the specified lateral branches, every $a = -N$ with $N \geq r$ has the germ (2.13).

Proof. On any compact set in the slit plane, choose L so large that $\Re(a + L) > 0$. In (2.9), the only powers that remain have $n \geq r$, hence their arguments avoid the principal logarithm cut. The tail is defined in the original half-plane. The rational coefficients have only possible poles at negative integers. Those in the slit domain have $N < r$, and Theorem 2.2 proves their removability. The formulas agree on overlaps by finite transport and coincide with the original function on a common open set. They therefore give the claimed jointly holomorphic continuation. $\square$

A distinction between principal continuation and all sheets. The early removable points cannot simply be deleted from the puncture set for every sheet. A positive circuit around $a = -N$ changes the chosen germ by

$$\Delta_N D_r(s; a) = (\exp(-2\pi i s) - 1) e_r(N; a) (a + N)^{-s}. \quad (2.16)$$

This follows by changing only $\log(a + N)$ in a finite transport formula with $L > N$. The rational coefficient $e_r(N; a)$ may have poles at earlier shifts. For instance, a circuit around -1 changes $D_1(s; a)$ by

$$(\exp(-2\pi i s) - 1) \frac{(a + 1)^{-s}}{a}.$$

Although the principal D_1 is regular at $a = 0$, this continued branch has a pole there for generic s . At $s = -m$, differentiating once in s gives the change $-2\pi i (a + 1)^m / a$. Thus the same issue occurs for spectral derivatives at the polynomial orders. Formula (2.15) is a statement about the principal branch. Arbitrary multivalued continuation can safely be made along paths avoiding all of $0, -1, -2, \dots$; it need not preserve the removability of every preimage of an earlier shift. This is a continuation qualification, not a counterexample to the source's half-plane theorem.

2.3 The first boundary and Arakawa–Kaneko values

For $j \geq 1$, write

$$\Xi_j(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{e^t - 1} \operatorname{Li}_j(1 - e^{-t}) dt, \quad \Re s > 0. \quad (2.17)$$

These are the classical Arakawa–Kaneko functions, continued entirely in s by Taylor subtraction at zero [8, 9]. In the descending convention, the multiple version $\Xi(j, \{1\}^{p-1}; s)$ is obtained by replacing the polylogarithm in (2.17) by $\text{Li}_{j,1,\dots,1}$. The mixed-spectral report proves the exact shift-jet identity

$$\partial_a^p D_q(s; a)|_{a=1} = (-1)^{q+p} p! \Xi(q+1, \{1\}^{p-1}; s), \quad p \geq 1. \quad (2.18)$$

In particular $\Xi_1(s) = s\zeta(s+1)$, with its removable value at $s = 0$ understood. This follows directly from $\text{Li}_1(1 - e^{-t}) = t$.

For a fixed positive depth r , set

$$E_h^{(r)} = e_h \left(1, \frac{1}{2}, \dots, \frac{1}{r}\right), \quad \prod_{j=1}^r \left(1 - \frac{z}{j}\right)^{-1} = \sum_{n \geq 0} h_n^{(r)} z^n, \quad (2.19)$$

$$c_r(z) = c_{r,r}(z) = \frac{1}{\prod_{j=1}^r (z-j)} = \frac{(-1)^r}{r!} \prod_{j=1}^r \left(1 - \frac{z}{j}\right)^{-1},$$

$$B_r(s) = (-1)^r \sum_{h=0}^{r-1} E_h^{(r)} \Xi_{r-h}(s). \quad (2.20)$$

The $h_n^{(r)}$ are complete homogeneous symmetric polynomials.

Theorem 2.4 (Exact first-boundary normalization). *For $r \geq 1$,*

$$\lim_{z \rightarrow 0} \{D_r(s; -r+z) - c_r(z) z^{-s}\} = B_r(s). \quad (2.21)$$

The expression inside braces extends holomorphically through $z = 0$. Convergence is locally uniform in s , and arbitrary fixed spectral derivatives can be applied to the identity. In particular,

$$\begin{aligned} B_1(s) &= -s\zeta(s+1), \\ B_2(s) &= \Xi_2(s) + \frac{3}{2}s\zeta(s+1), \\ B_3(s) &= -\Xi_3(s) - \frac{11}{6}\Xi_2(s) - s\zeta(s+1). \end{aligned} \quad (2.22)$$

Every Taylor coefficient of the holomorphic remainder in (2.21) is a finite combination of the multiple functions in (2.18), with rational coefficients.

Proof. Take $N = r$ in (2.12) and extract u^r . The finite terms with $n < r$ vanish because P_n has degree n . Thus the remainder is exactly

$$H_{r,r}(s; z) = [u^r] \left(1 + \frac{u}{z}\right) C_r(z, u) D(s, u; 1+z). \quad (2.23)$$

At $z = 0$, the term without u/z vanishes. The remaining term is $[u^r] u C_r(0, u) \partial_a D(s, u; 1)$. Now $C_r(0, u) = \sum_{h=0}^r (-1)^h E_h^{(r)} u^h$, while (2.18) at $p = 1$ gives $\partial_a D(s, u; 1) = \sum_{q \geq 0} (-1)^{q+1} \Xi_{q+1}(s) u^q$. Their finite coefficient product is precisely B_r . The holomorphy assertion is already contained in Theorem 2.2. For higher Taylor coefficients, expand $C_r(z, u)$ rationally at $z = 0$ and use

$$D(s, u; 1+z) = \sum_{p \geq 1} \sum_{q \geq 0} (-1)^{p+q} \Xi(q+1, \{1\}^{p-1}; s) u^q z^p.$$

At a fixed power of z in (2.23), only finitely many p, q occur. This also proves the final assertion with its normalization specified. $\square$

The same calculation gives every regular boundary constant. With $E_h^{(N)} = e_h(1, 1/2, \dots, 1/N)$ and $E_0^{(0)} = 1$,

$$H_{N,r}(s; 0) = \sum_{n=0}^{N-1} e_r(n; -N)(n - N)^{-s} + (-1)^r \sum_{h=0}^{\min(N, r-1)} E_h^{(N)} \Xi_{r-h}(s). \quad (2.24)$$

The second sum is empty for $r = 0$. This follows by evaluating the first line of (2.12) at zero and extracting the coefficient of u^r from $uC_N(0, u)\partial_a D(s, u; 1)$ in its second line. The finite powers retain their specified lateral logarithms. If $r > N$, the first sum is empty and the singular amplitude vanishes; the formula recovers the source's previously proved negative-shift values. If $r = N \geq 1$, it is (2.21).

The regular boundary term and a coordinate finite part. The subtraction in (2.21) retains the full analytic amplitude $c_r(z)$; it is not the same operation as discarding only the divergent terms of a Laurent expansion. For $p \geq 0$, take the finite part along the positive real coordinate z to mean the coefficient of $z^0(\log z)^0$ in the power-logarithm expansion. Then

$$\boxed{\text{FP}_{z=0} D_r(p; -r + z) = B_r(p) + \frac{(-1)^r}{r!} h_p^{(r)}}. \quad (2.25)$$

Indeed the constant contributed by $c_r(z)z^{-p}$ is $[z^p]c_r(z)$. On the other hand, for every $k \geq 1$,

$$\text{FP}_{z=0} \partial_s^k D_r(s; -r + z) \Big|_{s=p} = B_r^{(k)}(p), \quad p \geq 0, \quad (2.26)$$

because its singular amplitude has the factor $(-\log z)^k$. These formulas concern a fixed coordinate and a defined constant coefficient, so there is no implicit conversion between two regularization prescriptions.

For $m \geq 1$ and $k \geq 0$, there is an ordinary lateral limit,

$$\lim_{z \rightarrow 0} \partial_s^k D_r(s; -r + z) \Big|_{s=-m} = B_r^{(k)}(-m), \quad (2.27)$$

provided $\arg z$ stays in a bounded interval on the chosen branch. The reason is $z^m(\log z)^k \rightarrow 0$; no finite part is needed. For positive k , this continuous extension is not holomorphic, as the next subsection proves.

Explicit Stieltjes and Gamma boundary identities at depth one. Use the convention $\zeta(1+s) = s^{-1} + \sum_{j \geq 0} (-1)^j \gamma_j s^j / j!$. Then [10]

$$B_1(0) = -1, \quad B_1^{(k)}(0) = (-1)^k k \gamma_{k-1} \quad (k \geq 1). \quad (2.28)$$

Consequently, for example,

$$\begin{aligned} \lim_{z \rightarrow 0} \{D_1(0; -1 + z) - (z - 1)^{-1}\} &= -1, \\ \lim_{z \rightarrow 0} \{\partial_s D_1(0; -1 + z) + (z - 1)^{-1} \log z\} &= -\gamma, \\ \lim_{z \rightarrow 0} \{\partial_s^2 D_1(0; -1 + z) - (z - 1)^{-1} (\log z)^2\} &= 2\gamma_1. \end{aligned} \quad (2.29)$$

At the negative orders, for $m \geq 1$,

$$B_1^{(k)}(-m) = m \zeta^{(k)}(1 - m) - k \zeta^{(k-1)}(1 - m), \quad (2.30)$$

where the last term is omitted for $k = 0$. In particular,

$$B'_1(-1) = \frac{1}{2} - \frac{1}{2} \log(2\pi), \quad B'_1(-2) = \frac{1}{12} + 2\zeta'(-1).$$

The first evaluation uses the classical Lerch normalization of $\zeta'(0)$ [10]. These constants occur here as boundary values of a continued harmonic family, with the complete singular amplitude specified.

2.4 Normalized Lerch and polylogarithmic primitives

The exact singular amplitude also gives a finite antiderivative formula. This is an application of standard partial fractions and Lerch calculus [15], with the normalization needed at resonant spectral orders made explicit. Fix a logarithm on a slit punctured unit disk and a base point z_0 in that disk. For $1 \leq j \leq r$, put

$$A_{r,j} = \frac{(-1)^{r-j}}{(j-1)!(r-j)!}, \quad c_r(z) = \sum_{j=1}^r \frac{A_{r,j}}{z-j}, \quad (2.31)$$

$$P_j(s; z, z_0) = -\frac{1}{j} \left\{ z^{1-s} \Phi(z/j, 1, 1-s) - z_0^{1-s} \Phi(z_0/j, 1, 1-s) \right\}. \quad (2.32)$$

Here $\Phi(w, 1, v) = \sum_{n \geq 0} w^n / (n+v)$ for $|w| < 1$, continued meromorphically in v .

Proposition 2.5 (Entire normalized spectral primitive). *The difference P_j extends to an entire function of s , including the apparent resonances $s = 1, 2, \dots$, and*

$$P_j(s; z_0, z_0) = 0, \quad \frac{\partial}{\partial z} P_j(s; z, z_0) = \frac{z^{-s}}{z-j}. \quad (2.33)$$

Consequently a normalized primitive of the complete first-boundary germ is

$$\sum_{j=1}^r A_{r,j} P_j(s; z, z_0) + \int_{z_0}^z H_{r,r}(s; t) dt. \quad (2.34)$$

The remainder in this formula is holomorphic on the unit disk. Every fixed spectral derivative can be taken in the formula. At $s = 0$, all the differentiated singular primitives are finite ordinary-polylogarithm expressions: define

$$J_{j,k}(z) = -k! \sum_{h=0}^k \frac{(-\log z)^h}{h!} \text{Li}_{k+1-h}(z/j). \quad (2.35)$$

Then

$$\partial_s^k P_j(s; z, z_0) \Big|_{s=0} = J_{j,k}(z) - J_{j,k}(z_0), \quad J'_{j,k}(z) = \frac{(-\log z)^k}{z-j}. \quad (2.36)$$

Proof. The residues of c_r at its distinct simple poles give (2.31). Expanding the Lerch functions gives the normally convergent, already normalized series

$$P_j(s; z, z_0) = - \sum_{m \geq 1} \frac{z^{m-s} - z_0^{m-s}}{j^m (m-s)}. \quad (2.37)$$

For a term with $s = m$, interpret the quotient as $\log z - \log z_0$. Each term is entire in s . On compact spectral sets and compact subsets of the slit punctured disk, its tail is bounded by a geometric series times $1/m$. Hence the sum and all fixed spectral derivatives are locally normal. Equivalently, each unnormalized Lerch primitive in (2.32) has residue j^{-p} at $s = p \geq 1$,

independent of its endpoint; the difference cancels that residue. Differentiating (2.37) in z proves (2.33). Formula (2.23) shows that $H_{r,r}$ is holomorphic for $|z| < 1$, with its sole apparent pole at zero removed. Thus its integral is well defined independently of the path in that disk, and (2.34) follows. Finally, differentiating $z^{-s}/(m-s)$ at $s=0$ and summing the resulting powers m^{-q} gives (2.35); termwise differentiation, or the usual polylogarithm derivative rule, proves (2.36). $\square$

In particular,

$$J_{j,0}(z) = \log(1 - z/j), \quad J_{j,1}(z) = -\operatorname{Li}_2(z/j) - \log z \log(1 - z/j).$$

The sign in the second formula is fixed by $J'_{j,1}(z) = -\log z/(z-j)$. The base-point subtraction supplies the additive constant before continuation or spectral differentiation; separate unnormalized Lerch terms should not be evaluated at their resonant parameter poles.

2.5 Exact Taylor radii and the rank jump at integral orders

Theorem 2.6 (Exact shift Taylor radius). *Fix a real center $a_* > -r$ and a spectral order s_0 . The Taylor series in $a - a_*$ of $D_r(s_0; a)$ has radius*

$$\begin{cases} a_* + r, & s_0 \notin \{0, -1, -2, \dots\}, \\ \infty, & s_0 \in \{0, -1, -2, \dots\}. \end{cases}$$

For every $k \geq 1$ and every $s_0 \in \mathbb{C}$, the Taylor series of $\partial_s^k D_r(s_0; a)$ has radius exactly $a_ + r$. In particular, at the center $a_* = 1$, the lower bound $r + 1$ in [2] is exact outside the polynomial cases, and is exact for every positive spectral derivative even at those cases.*

Proof. The disk $|a - a_*| < a_* + r$ lies in the domain (2.15), so holomorphy gives the lower bound, uniformly on compact spectral sets. At $a = -r$, the singular amplitude is $c_r(z)z^{-s_0}(-\log z)^k$, with $c_r(0) = (-1)^r/r! \neq 0$; for $r = 0$ the same statement has $c_0 = 1$. When $k = 0$ and s_0 is not a nonpositive integer, z^{-s_0} has either a pole or nontrivial branching. It therefore cannot extend holomorphically to zero. For $k \geq 1$, its logarithmic factor prevents holomorphic extension for every s_0 . More explicitly, one circuit sends the germ to $e^{-2\pi i s_0} z^{-s_0}(-\log z - 2\pi i)^k$. If s_0 is nonintegral, the exponential factor is nontrivial; if s_0 is integral, the nonconstant logarithmic polynomial is not translation invariant. Positive integral s_0 at $k = 0$ instead gives a pole. The holomorphic remainder cannot cancel any of these obstructions. Hence the Taylor disk cannot be larger. At $s_0 = -m$, the finite Stirling formula below proves that $D_r(-m; a)$ is a polynomial and has infinite radius. $\square$

Corollary 2.7 (Functional independence of spectral jets). *If $s_0 \notin \{0, -1, -2, \dots\}$, every finite family*

$$\{\partial_s^k D_r(s_0; a) : 0 \leq r \leq R, 0 \leq k \leq K\}$$

is linearly independent over $\mathbb{C}$, even modulo entire functions of a . If $s_0 = -m$ with $m \geq 0$, the same assertion holds for the subfamily $1 \leq k \leq K$. This is independence of functions of the shift, not arithmetic independence of any special values.

Proof. Suppose a finite linear combination equals an entire function, and let r be the smallest depth with a nonzero coefficient. Every larger-depth term is holomorphic at $a = -r$ on the principal continuation. The singular germ of the depth- r terms is $c_r(z)z^{-s_0}P(-\log z)$ for a nonzero polynomial P . The monodromy argument from Theorem 2.6 shows that this cannot be holomorphic, except possibly when s_0 is a nonpositive integer and P is constant. That exception is excluded by the stated subfamily. Thus all coefficients at this depth vanish, a contradiction. Induction gives the claim. $\square$

Let $\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \}$ denote a Stirling number of the second kind and let $(a-1)^{\underline{j}} = (a-1) \cdots (a-j)$. The known finite formula [3, 2] is

$$D_r(-m; a) = \sum_{j=1}^{m+1} \left\{ \begin{smallmatrix} m+1 \\ j \end{smallmatrix} \right\} (a-1)^{\underline{j}} \left(-\frac{1}{j} \right)^{r+1}. \quad (2.38)$$

Proposition 2.8 (Exact finite rank and minimal depth recurrence). *For $m \geq 0$, the functions $\{D_r(-m; a) : r \geq 0\}$ span exactly the $(m+1)$ -dimensional space of polynomials of degree at most $m+1$ vanishing at $a=1$. The first $m+1$ depths form a basis. If E shifts depth, $ED_r = D_{r+1}$, then*

$$\prod_{j=1}^{m+1} \left(E + \frac{1}{j} \right) D_r(-m; a) = 0, \quad r \geq 0. \quad (2.39)$$

Its order $m+1$ is minimal among nonzero constant-coefficient depth recurrences valid as identities in a . For any fixed $k \geq 1$, the family $\{\partial_s^k D_r(-m; a) : r \geq 0\}$ is infinite dimensional and satisfies no nonzero constant-coefficient depth recurrence. The latter statement also holds at nonexceptional s_0 with $k=0$.

Proof. The polynomials $\left\{ \begin{smallmatrix} m+1 \\ j \end{smallmatrix} \right\} (a-1)^{\underline{j}}$, $1 \leq j \leq m+1$, have distinct degrees, nonzero leading coefficients, and vanish at one. They form a basis of the stated space. The coefficient matrix for depths $0, \dots, m$ in (2.38) has entries λ_j^{r+1} , where $\lambda_j = -1/j$. Its determinant is a nonzero product of the λ_j and a Vandermonde determinant. This proves the rank and basis assertions. Each component in (2.38) is annihilated by $E - \lambda_j$, which proves (2.39). Conversely, independence of those polynomial components forces an annihilating polynomial Q to vanish at every λ_j . Its degree is therefore at least $m+1$. Finally any recurrence for a positive spectral derivative would give a finite constant linear relation among distinct depths, contradicting Corollary 2.7. $\square$

For example, at $s=0$ one has $D_r(0; a) = (-1)^{r+1}(a-1)$ and $(E+1)D_r = 0$. At $s=-1$ the minimal recurrence is $D_{r+2} + \frac{3}{2}D_{r+1} + \frac{1}{2}D_r = 0$. A single spectral differentiation removes these finite-rank relations. This failure is detected by exact logarithmic germs, without any numerical independence assumption.

2.6 Audit and reproducibility

The source's holomorphy domain and Taylor lower bound were correct; the new statements determine their actual boundary behavior. The regular boundary term B_r and the coordinate finite part in (2.25) must retain their different definitions. Likewise, principal-sheet removability must not be promoted to removability on all sheets: (2.16) supplies an explicit obstruction.

The accompanying program `code/verify_shift_germs.py` checks the finite recurrence and residue algebra exactly with rational arithmetic, compares the Newton series with the independently convergent defining harmonic sums at complex parameters, checks finite transport at negative shifts, and compares Cauchy-extracted boundary constants with Arakawa–Kaneko integrals and ordinary zeta derivatives. The numerical checks are finite-precision diagnostics with separately recorded truncations; they are not interval certificates or replacements for the normal-convergence and continuation arguments above. The short companion `code/verify_shift_primitives.py` checks the partial-fraction amplitudes and the polylogarithmic primitive derivatives with exact symbolic arithmetic.

3 The exact space of mixed polygamma moments

The mixed-spectral report asks which individual polygamma moments can be separated from its two-parameter Gamma identity, with explicit first targets at total orders six and eight [2].

This section answers that question for the stated family at every weight. There are two different dimensions to compute: the dimension of its formal polynomial rows, and the dimension of the functions obtained after summation. The former grows with the weight; the latter is exactly three from weight six onward. Their difference produces an infinite family of identically vanishing, absolutely convergent polygamma sums.

3.1 Normalization and the underlying Gamma identity

For $\Re a > 0$, set $x_n = n + a/2$, $A = a - 1$, and

$$L_k(n; a) = (-1)^k \psi^{(k-1)}(n + a) - \psi^{(k-1)}(n + 1), \quad k \geq 2. \quad (3.1)$$

These are ordinary functions, and all products below are algebraic products, without complex conjugation. Write

$$Q_n(p, q; a) = \frac{\Gamma(n + a)\Gamma(n + 1 + p)\Gamma(n + 1 + q)\Gamma(n + a - p - q)}{\Gamma(n + 1)\Gamma(n + a - p)\Gamma(n + a - q)\Gamma(n + 1 + p + q)}, \quad (3.2)$$

$$b_a(p, q) = \psi(a/2) + \frac{1}{2}\{\psi(2) - \psi(1 + p) - \psi(1 + q) - \psi(a - p - q)\}. \quad (3.3)$$

Lemma 3.1 (The double-zero Gamma sum: antecedent). *For sufficiently small p, q , locally uniformly with $\Re a > 0$,*

$$\sum_{n \geq 0} \left\{ x_n(Q_n(p, q; a) - 1) + \frac{(A - p - q)pq}{x_n} \right\} = -(A - p - q)pq b_a(p, q). \quad (3.4)$$

Every fixed parameter derivative can be taken in the displayed bracket.

Proof. This is the double-zero identity proved in [2]. We give the residue calculation to fix its normalization. In the Rogers–Dougall ${}_5F_4(1)$ sum [13, 16.4.9], set $b = 1 + p$, $c = 1 + q$, and $d_0 = a - 1 - p - q$. Introduce the excess parameter v ; the centered summand and its sum are

$$r_n(v) = x_n \frac{\Gamma(n + a)\Gamma(n + b)\Gamma(n + c)\Gamma(n + d_0 - v)}{\Gamma(n + 1)\Gamma(n + 1 + a - b)\Gamma(n + 1 + a - c)\Gamma(n + b + c + v)},$$

$$G(v) = \frac{\Gamma(b)\Gamma(c)\Gamma(d_0 - v)\Gamma(v)}{2\Gamma(d_0)\Gamma(b + v)\Gamma(c + v)}.$$

The centered Stirling expansion [12] is $r_n(v) = x_n^{-1-2v}(1 + C_1(v)x_n^{-2} + O(x_n^{-4}))$. If $\eta = a/2$ and

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (\eta, 1 + p - \eta, 1 + q - \eta, \eta - 1 - p - q - v),$$

then $C_1(v) = -\frac{1}{3} \sum_i B_3(\alpha_i)$. Subtract the first two centered terms. Normal convergence continues the resulting sum from $\Re v > 0$ to a neighborhood of $v = -1$. At that point $r_n(-1) = x_n Q_n$ and

$$\rho = C_1(-1) = -d_0 pq, \quad C'_1(-1) = B_2(\eta - p - q), \quad \zeta(-1, \eta) = -\frac{1}{2} B_2(\eta).$$

The finite part of $G(v)$ is

$$\frac{\rho}{2} \{\psi(2) - \psi(a - p - q) - \psi(p) - \psi(q)\}.$$

The polar Hurwitz counterterm contributes $C'_1(-1)/2 - \rho\psi(\eta)$. Finally,

$$\frac{1}{2} C'_1(-1) + \zeta(-1, \eta) = -\frac{1}{2} d_0(p + q).$$

Using $\psi(p) = \psi(1 + p) - 1/p$ and its q counterpart cancels these last terms and gives (3.4). The calculation first applies for $a > 1$ and generic small p, q . The bracket is uniformly $O(n^{-3})$ on compact parameter sets, and its Gamma arguments avoid poles in a neighborhood of every $(a, 0, 0)$ with $\Re a > 0$. Analytic continuation therefore gives the asserted domain. Cauchy's formula on a slightly larger parameter polydisc justifies every fixed derivative and includes the loci $p = 0$ or $q = 0$. $\square$

Let $L_2, L_3, \dots$ temporarily be independent indeterminates of weights $2, 3, \dots$. Define

$$\mathcal{B}_{r,s} = r!s![p^r q^s] \exp \left(\sum_{k \geq 2} \frac{L_k}{k!} \{(p+q)^k - p^k - q^k\} \right), \quad r, s \geq 1. \quad (3.5)$$

Only finitely many coefficients enter each polynomial. Taylor expansion of the logarithm of (3.2) gives

$$\log Q_n(p, q; a) = \sum_{i,j \geq 1} L_{i+j}(n; a) \frac{p^i q^j}{i!j!}.$$

Thus (3.5) specializes to the mixed derivative of Q_n at the origin. We retain the distinction between an indeterminate L_k and its specialization (3.1) whenever dimensions are at issue.

3.2 Formal rank at every weight

Let $\mathcal{P}_w$ be the rational vector space spanned by all monomials of weight w in $L_2, L_3, \dots$. Let $p(w)$ be the ordinary integer partition number, with $p(0) = 1$. Its dimension is $p(w) - p(w-1)$, since these are exactly the partitions without a part one. Set

$$\mathcal{V}_w = \text{span}_{\mathbb{Q}} \{\mathcal{B}_{r,w-r} : 1 \leq r \leq \lfloor w/2 \rfloor\}.$$

The symmetry exchanging the two Gamma parameters has already been included: $\mathcal{B}_{r,s} = \mathcal{B}_{s,r}$. The elementary linear row is already present, since $\mathcal{B}_{1,w-1} = L_w$.

Theorem 3.2 (Complete formal rank and finite membership test). *For every integer $w \geq 2$,*

$$\dim_{\mathbb{Q}} \mathcal{V}_w = \lfloor w/2 \rfloor, \quad \text{codim}_{\mathcal{P}_w} \mathcal{V}_w = p(w) - p(w-1) - \lfloor w/2 \rfloor. \quad (3.6)$$

For a monomial $L_{\lambda} = \prod_{k \geq 2} L_k^{\nu_k}$, $\sum k\nu_k = w$, its coefficient in row r is exactly

$$M_w(r, \lambda) = \frac{r!(w-r)!}{\prod_k \nu_k! (k!)^{\nu_k}} [t^r] \prod_{k \geq 2} \{(1+t)^k - 1 - t^k\}^{\nu_k}. \quad (3.7)$$

Consequently rational row reduction of this explicit finite matrix decides membership in the stated moment space at any prescribed weight.

Proof. Expanding the exponential in (3.5), and then setting $q = 1$, proves (3.7). A monomial with ℓ factors requires at least ℓ powers of both p and q , because every term in the exponent is divisible by pq . Hence a row with first index r contains no monomial with more than r factors.

For $1 \leq j \leq \lfloor w/2 \rfloor$, select the monomial

$$U_j = L_2^{j-1} L_{w-2j+2}.$$

It has exactly j factors, including when it is L_2^j . The coefficient of U_j vanishes in every row $r < j$. In row $r = j$ it is strictly positive: choose the linear term of each factor $(1+t)^k - 1 - t^k$, whose coefficients are positive. The square minor formed by these columns is triangular with nonzero diagonal. The rows are therefore independent. There are exactly $\lfloor w/2 \rfloor$ of them, which proves both dimension formulas and the membership claim. $\square$

Remark 3.3 (The scope of the obstruction). Nonmembership in $\mathcal{V}_w$ means that these particular Gamma derivative rows and their parameter-exchange symmetry do not isolate the desired polynomial. It does not imply that its sum lacks a closed form, nor that evaluated special values are arithmetically independent. Differentiating in a , shifting summation indices, or using a different hypergeometric identity can supply additional relations outside this space.

3.3 The requested order-six and order-eight elimination matrices

At $w = 6$ order the columns as $(L_6, L_2L_4, L_3^2, L_2^3)$ and the rows as $(1, 5), (2, 4), (3, 3)$. Then

$$M_6 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 8 & 6 & 0 \\ 1 & 9 & 9 & 6 \end{pmatrix}, \quad E_6 = \begin{pmatrix} 1 & 0 & 0 \\ -1/6 & 1/2 & -1/3 \\ 1/18 & -1/2 & 4/9 \end{pmatrix}, \quad (3.8)$$

and exact multiplication gives

$$E_6 M_6 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 8/3 \end{pmatrix}. \quad (3.9)$$

Thus the combinations $L_2L_4 - 2L_2^3$ and $L_3^2 + \frac{8}{3}L_2^3$ are evaluated by the family. None of the three nonlinear monomials is individually in its row space. The column vector $(0, 2, -8/3, 1)^T$ annihilates every row and has a nonzero entry in each nonlinear position, giving a short exact certificate of this assertion.

At $w = 8$, take the columns

$$(L_8, L_2L_6, L_3L_5, L_4^2, L_2^2L_4, L_2L_3^2, L_2^4)$$

and the rows $(1, 7), (2, 6), (3, 5), (4, 4)$. The matrix is

$$M_8 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 12 & 30 & 20 & 0 & 0 & 0 \\ 1 & 15 & 45 & 30 & 60 & 90 & 0 \\ 1 & 16 & 48 & 34 & 72 & 144 & 24 \end{pmatrix}. \quad (3.10)$$

An explicit elimination matrix and its result are

$$E_8 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1/6 & 1/2 & -1/3 & 0 \\ 1/90 & -1/6 & 22/45 & -1/3 \\ 1/30 & 0 & -8/15 & 1/2 \end{pmatrix},$$

$$E_8 M_8 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -20 & -30 & 0 \\ 0 & 0 & 1 & 0 & 16/3 & -4 & -8 \\ 0 & 0 & 0 & 1 & 4 & 24 & 12 \end{pmatrix}. \quad (3.11)$$

The three free columns give the following basis of the annihilator:

$$\begin{pmatrix} 0 \\ 20 \\ -16/3 \\ -4 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 30 \\ 4 \\ -24 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 8 \\ -12 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (3.12)$$

Every nonlinear column has a nonzero coordinate in at least one of these annihilators. Therefore L_8 is the only individual monomial in this weight that the specified row space isolates. Equations (3.8)–(3.12) answer the requested finite elimination problem with exact rational certificates.

3.4 Summation reduces the image to three functions

For $w \geq 6$, write

$$T_w(a) = (a-1)\psi^{(w-2)}(a) + (w-2)\psi^{(w-3)}(a), \quad \epsilon_w = (-1)^w, \quad (3.13)$$

and let $S_r(a)$ denote $\sum_{n \geq 0} x_n \mathcal{B}_{r,w-r}(n; a)$. No subtraction is needed at these weights. More generally, every monomial of weight $w \geq 4$ has an absolutely convergent weighted sum: the polygamma series gives $L_{2j} = O(n^{-2j})$ and $L_{2j+1} = O(n^{-2j})$, locally uniformly in a . The smallest resulting decay exponent is at least four, before multiplication by x_n .

Theorem 3.4 (Exact summation image and all its zero rows). *For $w \geq 6$ the moment functions are*

$$S_1(a) = \frac{w-1}{2} \left\{ \epsilon_w T_w(a) + (a-1)\psi^{(w-2)}(1) - (w-2)\psi^{(w-3)}(1) \right\}, \quad (3.14)$$

$$S_2(a) = \epsilon_w (w-2) T_w(a) - (w-2)\psi^{(w-3)}(1), \quad (3.15)$$

$$S_r(a) = \frac{\epsilon_w r(w-r)}{2} T_w(a), \quad 3 \leq r \leq \lfloor w/2 \rfloor. \quad (3.16)$$

The linear map from $\mathcal{V}_w \otimes_{\mathbb{Q}} \mathbb{C}$ to holomorphic functions of a obtained by weighted summation has rank exactly three. Its kernel has dimension $\lfloor w/2 \rfloor - 3$, and a rational basis is

$$\frac{\mathcal{B}_{r,w-r}}{r(w-r)} - \frac{\mathcal{B}_{3,w-3}}{3(w-3)}, \quad 4 \leq r \leq \lfloor w/2 \rfloor. \quad (3.17)$$

Every one of these polynomials has identically zero weighted sum for $\Re a > 0$.

Proof. Put $b_{i,j} = \partial_p^i \partial_q^j b_a(0,0)$. For $i+j = k \geq 1$,

$$b_{i,j} = -\frac{1}{2} \left\{ \mathbf{1}_{j=0} \psi^{(i)}(1) + \mathbf{1}_{i=0} \psi^{(j)}(1) + (-1)^k \psi^{(k)}(a) \right\}.$$

Differentiating (3.4) gives

$$S_r(a) = -r(w-r) \{ A b_{r-1,w-r-1} - (r-1) b_{r-2,w-r-1} - (w-r-1) b_{r-1,w-r-2} \}.$$

Terms with a negative b -index are omitted; their prefactors are zero. The counterterm has total degree only three, and therefore contributes nothing here. The cases $r=1$, $r=2$, and $r \geq 3$ are precisely (3.14)–(3.16).

To prove exact rank, not merely an upper bound, let a tend to positive infinity. The standard polygamma asymptotic yields

$$T_w(a) = \epsilon_w (w-4)! a^{-(w-3)} + O(a^{-(w-2)}).$$

The first row has a nonzero linear term, the second has a nonzero constant limit, and the third is a nonzero function tending to zero. In any constant linear relation, these three successive behaviors force the coefficients of S_1 , S_2 , and S_3 to vanish. The image rank is exactly three. The displayed kernel rows are independent in the formal basis of Theorem 3.2; their number equals the nullity, so they form the entire kernel. This proves the claim over $\mathbb{C}$ and hence also its rational version. $\square$

For example, the two evaluable order-six combinations in (3.9) give

$$\sum_{n \geq 0} x_n (L_2 L_4 - 2L_2^3) = \frac{T_6(a)}{12} + 10(a-1)\zeta(5) - 2\zeta(4), \quad (3.18)$$

$$\sum_{n \geq 0} x_n \left(L_3^2 + \frac{8}{3} L_2^3 \right) = \frac{5T_6(a)}{36} - \frac{10(a-1)}{3} \zeta(5) + \frac{26}{3} \zeta(4). \quad (3.19)$$

Here and below every L_k in a summand means $L_k(n; a)$.

Corollary 3.5 (A quartic polygamma identity). *For $\Re a > 0$,*

$$\sum_{n \geq 0} x_n \{L_8 - 30L_4^2 - 120L_2^2L_4 - 720L_2L_3^2 - 360L_2^4\} = 0. \quad (3.20)$$

Equivalently,

$$\begin{aligned} \sum_{n \geq 0} x_n \{L_4^2 + 4L_2^2L_4 + 24L_2L_3^2 + 12L_2^4\} \\ = \frac{7}{60} \{T_8(a) + (a-1)\psi^{(6)}(1) - 6\psi^{(5)}(1)\}. \end{aligned} \quad (3.21)$$

Proof. The polynomial in (3.20) is exactly $16\mathcal{B}_{3,5} - 15\mathcal{B}_{4,4}$, by (3.10). Its sum is zero by (3.16). Move the linear term to the other side and apply (3.14). All the individual sums converge, so this rearrangement needs no regularized interpretation. $\square$

3.5 A harmonic-tail and multiple-polylogarithm consequence

For $k \geq 1$, define the centered cubic tail

$$C_3(k) = 2\{\zeta(3) - H_{k-1}^{(3)}\} - k^{-3} = k^{-3} + 2 \sum_{j>k} j^{-3}.$$

The next identity is a consequence of the order-eight null row, rather than an independently asserted new evaluation in the classical Euler-sum literature.

Corollary 3.6 (A reciprocal-weight cubic-tail square). *One has the ordinary convergent identities*

$$\sum_{k=1}^{\infty} \frac{C_3(k)^2}{k} = 2\zeta(7), \quad (3.22)$$

$$\text{Li}_{3,4}(1) + 2 \text{Li}_{3,3,1}(1) + \text{Li}_{6,1}(1) = \frac{1}{4}\zeta(7). \quad (3.23)$$

The multiple polylogarithms use strictly decreasing positive indices.

Proof. At $a = 2$, set $k = n + 1$. The polygamma functional equation gives

$$x_n = k, \quad L_2 = -k^{-2}, \quad L_4 = -6k^{-4}, \quad L_8 = -5040k^{-8}, \quad L_3 = 2C_3(k).$$

Substituting in (3.20) yields $-5760\zeta(7) + 2880 \sum_k C_3(k)^2/k = 0$, proving (3.22). Since all terms in the defining tail are positive, expansion and rearrangement are justified by monotone convergence (or directly by absolute convergence). In decreasing-index notation that expansion is

$$\sum_k \frac{C_3(k)^2}{k} = \zeta(7) + 4\zeta(3,4) + 8\zeta(3,3,1) + 4\zeta(6,1).$$

Combining the two expressions proves (3.23). $\square$

The rank theorem explains both the usefulness and the limitations of the two-parameter family. It supplies many exact nonlinear combinations and an increasing number of zero-sum relations, while the explicit annihilators at weights six and eight explain why an individual unseparated moment still requires additional information.

4 A finite Euler-sum space for every centered harmonic moment

The recent raw-power report proves that the centered series

$$F_p(s, a) = \sum_{n=0}^{\infty} \frac{H_n^p}{(n+a)^s}, \quad H_n = \sum_{j=1}^n \frac{1}{j}, \quad H_0 = 0,$$

is regular at $s = 0, -2, -4, \dots$ when $a = 1/2$. Its Question 10 asks for a generator that avoids listing every composition of p , with the complete sequence $F_p(-2, 1/2)$ as a first target [5]. We give such a generator for *every* nonnegative even moment. More strongly, for fixed p all these values lie in one finite rational span of convergent Euler sums, independent of the moment order. These are exact analytic identities; no assertion of arithmetic independence or literature-wide priority is made. They concern values at regular spectral points, not derivatives of a formula known only at those points.

4.1 An exponent parameter with the required joint regularity

Introduce the auxiliary exponent family

$$\mathcal{F}(s, u) = \sum_{n=0}^{\infty} \frac{\exp(uH_n)}{(n+1/2)^s}, \quad \Re(s-u) > 1. \quad (4.1)$$

Positive bases have their real logarithms. This normally convergent series is jointly holomorphic in its initial domain. At $u = 0$ its p th derivative is $F_p(s, 1/2)$ for $p \geq 1$; the zeroth derivative is $\zeta(s, 1/2)$. Let $B_j(x)$ denote the Bernoulli polynomials, and put

$$b_j = -\frac{B_{2j}(1/2)}{2j}, \quad c_j(u) = [z^j] \exp\left(u \sum_{\ell \geq 1} b_\ell z^\ell\right). \quad (4.2)$$

The latter definition is formal: each c_j is a finite polynomial. No convergence of the infinite Bernoulli series is required.

Lemma 4.1 (Joint continuation and the centered divisor pattern). *The family $\mathcal{F}$ has a joint meromorphic continuation to $\mathbb{C}^2$, with possible polar hyperplanes*

$$s - u = 1 - 2j, \quad j = 0, 1, 2, \dots \quad (4.3)$$

In particular it is jointly holomorphic near every $(-2r, 0)$, $r \geq 0$. The function

$$V_r(u) = \mathcal{F}(-2r, u) = \sum_{p \geq 1} F_p(-2r, 1/2) \frac{u^p}{p!} \quad (4.4)$$

is holomorphic for $|u| < 1$.

Proof. With $x = n + 1/2$, Euler–Maclaurin gives, for every fixed J ,

$$H_n = \log x + \gamma + \sum_{j=1}^J b_j x^{-2j} + O(x^{-2J-2}).$$

This is the centered shifted-digamma expansion used in [5]. Exponentiating a finite truncation gives

$$\exp(uH_n) = e^{\gamma u} x^u \left(\sum_{j=0}^J c_j(u) x^{-2j} + O_K(x^{-2J-2}) \right), \quad (4.5)$$

uniformly for u in a compact set K . Fixed u derivatives have analogous estimates with powers of $\log x$. Subtract these finitely many terms in (4.1), starting at any fixed positive index. The remainder is normally convergent when $\Re(s - u) > -2J - 1$, and the subtracted terms are $e^{\gamma u} c_j(u) \zeta(s - u + 2j, N + 1/2)$. The sole pole of each Hurwitz function is at its first argument 1 [10]. Arbitrarily large J prove the stated joint continuation. None of the hyperplanes (4.3) passes through $(-2r, 0)$. Joint holomorphy, rather than a formal interchange at a singular point, therefore identifies all the coefficients in (4.4). Its constant coefficient is $\zeta(-2r, 1/2) = 0$. $\square$

4.2 A triangular generator from ordinary telescoping

For $k \geq 2$ define the normally convergent functions

$$E_k(u) = \sum_{n \geq 1} \frac{\exp(uH_{n-1})}{n^k}, \quad \Re u < k - 1, \quad (4.6)$$

$$B(u) = 1 + \sum_{k \geq 2} \frac{u^k}{k!} E_k(u), \quad \Re u < 1. \quad (4.7)$$

The letter B without an index in (4.7) is an auxiliary function, not a Bernoulli polynomial. Its coefficients involve the ordinary convergent Euler sums

$$E_{q,k} = \sum_{n \geq 1} \frac{H_{n-1}^q}{n^k}, \quad q \geq 0, \quad k \geq 2; \quad E_k(u) = \sum_{q \geq 0} E_{q,k} \frac{u^q}{q!}. \quad (4.8)$$

The Taylor identity in (4.8) is used near $u = 0$. Here $H_{n-1}^0 = 1$, including at $n = 1$.

Theorem 4.2 (A finite triangular generator at every centered moment). *For $m = 0, 1, \dots$, define $G_m(u)$ recursively by*

$$\begin{aligned} (u + m + 1)G_m(u) &= \sum_{j=0}^{m-1} \left\{ (-1)^{m+1-j} \binom{m+1}{j} - \frac{u^{m+1-j}}{(m+1-j)!} \right\} G_j(u) \\ &\quad + \frac{u^{m+1}}{(m+2)!} B(u) - \sum_{k \geq 2} \frac{u^{k+m+1}}{(k+m+1)!} E_k(u). \end{aligned} \quad (4.9)$$

The first sum is empty at $m = 0$. All functions are holomorphic on $|u| < 1$, and

$$V_r(u) = \sum_{j=0}^{2r} \binom{2r}{j} \left(-\frac{1}{2}\right)^{2r-j} G_j(u). \quad (4.10)$$

Thus $2r + 1$ triangular steps determine the exponential generating function for all $F_p(-2r, 1/2)$. For a specified p , only finitely many coefficients of finitely many E_k are needed.

Proof. Put $g_n(u) = \exp(uH_{n-1})$. It obeys the exact discrete equation

$$g_{n+1}(u) = e^{u/n} g_n(u).$$

For $\Re u < 0$, telescoping and then an absolutely convergent exponential expansion give

$$-1 = \sum_{n \geq 1} (g_{n+1} - g_n) = u \sum_{n \geq 1} \frac{g_n}{n} + \sum_{k \geq 2} \frac{u^k}{k!} E_k(u).$$

Consequently the continued order-one sum is exactly

$$\sum_{n \geq 1} \frac{g_n(u)}{n} = -\frac{B(u)}{u}. \quad (4.11)$$

This equality initially concerns ordinary sums in $\Re u < 0$.

For a fixed $m \geq 0$, work first in $\Re u < -m - 1$, where all the following sums converge absolutely. Multiplying the discrete equation by n^{m+1} , summing to a finite cutoff N , and only then shifting an index yields the following formulas after $N \rightarrow \infty$

$$\begin{aligned} \sum_{n \geq 1} n^{m+1}(g_{n+1} - g_n) &= \sum_{j=0}^m (-1)^{m+1-j} \binom{m+1}{j} \sum_{n \geq 1} n^j g_n, \\ \sum_{n \geq 1} n^{m+1}(g_{n+1} - g_n) &= \sum_{k \geq 1} \frac{u^k}{k!} \sum_{n \geq 1} \frac{g_n}{n^{k-m-1}}. \end{aligned}$$

The upper-end term is $N^{m+1}g_{N+1} = O(N^{m+1+\Re u})$ and vanishes in the stated half-plane. There is no lower-end boundary term because the polynomial being shifted has positive degree. This finite-cutoff argument never separates a difference into two possibly divergent infinite sums. In the second line separate $k = 1$, $2 \leq k \leq m+1$, $k = m+2$, and $k \geq m+3$. Using (4.11) for the order-one term and moving the two copies of $\sum n^m g_n$ to the left gives (4.9). In particular its G_m is the meromorphic continuation of the ordinary sum $\sum_{n \geq 1} n^m \exp(uH_{n-1})$ from $\Re u < -m - 1$.

Here are the convergence details for the continuation. On a compact subset of $\Re u < 1$, choose $U < 1$ with $\Re u \leq U$. The two-sided estimate $H_{n-1} = \log n + O(1)$ gives $|g_n(u)| \leq Cn^U$, uniformly, including when $\Re u$ is negative. Hence all $E_k(u)$, $k \geq 2$, are bounded by a common summable majorant, and the factorial denominators make both infinite k sums normally convergent. Fixed derivatives in u are justified on slightly larger compact subsets, or directly by inserting powers of H_{n-1} . The recurrence therefore continues G_m meromorphically to $\Re u < 1$, with possible poles only at $-1, -2, \dots, -m - 1$. In particular it is holomorphic on $|u| < 1$.

Finally, for $\Re u < -2r - 1$, an ordinary change of index in the absolutely convergent defining series gives

$$\mathcal{F}(-2r, u) = \sum_{n \geq 1} (n - 1/2)^{2r} g_n(u) = \sum_{j=0}^{2r} \binom{2r}{j} (-1/2)^{2r-j} G_j(u).$$

Meromorphic continuation in u proves (4.10); Lemma 4.1 identifies its Taylor coefficients with the claimed regular spectral values. $\square$

Remark 4.3 (Why the auxiliary values cannot be specialized in the other order). Induction in (4.9) gives $G_m(0) = 0$ for every m . For example $G_1(0) = 0$, whereas $\zeta(-1) = -1/12$. This is consistent: G_m first fixes the spectral argument at $-m$ and then continues the exponent u . The uncentered family has polar hyperplanes $s - u = 1 - j$. Even a single such divisor can prevent joint holomorphy when its numerator vanishes at the specialization point. For example its local subtraction includes

$$e^{\gamma u} \left(\frac{u^2}{8} - \frac{u}{12} \right) \zeta(s - u + 2).$$

At $s = -1$ and then $u \rightarrow 0$, this contributes $1/12$; at $u = 0$ first, it contributes zero. It is therefore invalid to replace $G_m(0)$ by the separately specialized zeta value. The centered combination (4.10) is justified by joint holomorphy, which was proved before coefficient extraction.

4.3 A finite arithmetic space independent of the moment order

Theorem 4.4 (Uniform finite Euler-sum span). *For $p \geq 1$, set*

$$\mathcal{E}_{p-1} = \text{span}_{\mathbb{Q}}(\{1\} \cup \{E_{q,k} : q \geq 0, k \geq 2, q + k \leq p - 1\}). \quad (4.12)$$

Then

$$\boxed{F_p(-2r, 1/2) \in \mathcal{E}_{p-1} \quad \text{for every } r \geq 0.} \quad (4.13)$$

Every value has a finite multiple-zeta expression using weights at most $p - 1$, with no Euler constant required. In particular every centered even-negative moment of H_n or H_n^2 is rational. For $p \geq 2$, the displayed spanning list has at most $1 + \binom{p-1}{2}$ elements; this is a spanning bound, not a claim that the periods in the list are independent.

Proof. The recurrence is linear in the functions $E_k(u)$, and all rational coefficients being expanded are regular at $u = 0$. An occurrence of E_k first has a factor u^{k+m+1} , either in the last sum of (4.9) or through B in its preceding term. Its exponent is at least $k + 1$. Induction on m therefore shows that the coefficient of u^p in any G_m , and hence in any V_r , is a rational linear combination of 1 and $E_{q,k}$ with $q + k \leq p - 1$.

For completeness, the finite multiple-zeta reduction is

$$E_{q,k} = \sum_{(r_1, \dots, r_d) \text{ a composition of } q} \frac{q!}{r_1! \cdots r_d!} \zeta(k, r_1, \dots, r_d). \quad (4.14)$$

At $q = 0$ the empty composition gives $E_{0,k} = \zeta(k)$. To prove the formula, expand H_{n-1}^q , order its distinct indices, and group equal indices into blocks of sizes $r_1, \dots, r_d$. The displayed multinomial is their multiplicity. Each resulting MZV converges because its first index is $k \geq 2$, and its weight is $q + k$. Formula (4.14) is an optional final conversion: it is not necessary for computing the generator itself. $\square$

This uniform space is smaller information than an unspecified multiple-zeta algebra at each separate moment. Increasing r changes rational coefficients in one fixed finite list; it does not require Euler sums of increasing weight.

4.4 Two all-moment Bernoulli identities

Corollary 4.5 (The first two powers at every centered even moment). *For every $r \geq 0$,*

$$F_1(-2r, 1/2) = \frac{1}{2} B_{2r}(1/2), \quad (4.15)$$

$$F_2(-2r, 1/2) = - \sum_{j=0}^r \binom{2r}{2j} \frac{B_{2j} B_{2r-2j}(1/2)}{2j+1}. \quad (4.16)$$

Proof. The classical generating functions needed here are particularly short. Write $L(t) = -\log(1 - e^{-t})$, and define $A_p(t) = (1 - e^{-t}) \sum_{n \geq 0} H_n^p e^{-nt}$. Then $A_1 = L$ and $A_2 = L^2 + \text{Li}_2(e^{-t})$. Consequently

$$A_2'(t) = -\coth(t/2)L(t).$$

The odd powers in the local expansion of L consist only of $t/2$:

$$L(t) = -\log t + \frac{t}{2} - \log \left(\frac{\sinh(t/2)}{t/2} \right).$$

Here “odd powers” refers to the power-log expansion at $t = 0$, with no branch continuation to negative t being assumed. It follows that the odd-power part of A_2 is

$$- \int_0^t \frac{v}{2} \coth(v/2) dv = - \sum_{j \geq 0} \frac{B_{2j}}{(2j)!(2j+1)} t^{2j+1}.$$

The centered Mellin kernel is $A_p(t)/(2 \sinh(t/2))$. Its log-free coefficient of t^{2r} , multiplied by $(2r)!$, is $F_p(-2r, 1/2)$. Indeed that coefficient gives the simple Mellin pole at $s = -2r$, and the simple

zero of $1/\Gamma(s)$ cancels it; all holomorphic Mellin terms vanish at the specialization. Regularity proved in Lemma 4.1 eliminates higher logarithmic poles. Finally use

$$\frac{1}{2 \sinh(t/2)} = \sum_{j \geq 0} \frac{B_{2j}(1/2)}{(2j)!} t^{2j-1}.$$

Multiplying the two convergent local series gives (4.15) and (4.16). $\square$

4.5 A sufficient parity class for generalized harmonic products

There is also a direct extension toward Question 11 of [5]. Define $H_n^{(q)} = \sum_{k=1}^n k^{-q}$ and $U_q(s) = \sum_{n \geq 0} H_n^{(q)}(n+1/2)^{-s}$.

Corollary 4.6 (Odd harmonic orders and finite ordinary-zeta values). *Let P be any polynomial in finitely many variables. The Dirichlet series with numerator*

$$P(H_n, H_n^{(3)}, H_n^{(5)}, \dots, H_n^{(2d+1)})$$

is holomorphic at every $s = -2r$, $r \geq 0$. This is a sufficient class of numerator polynomials, not a necessity classification. For a single positive odd integer q , the exact values are

$$U_q(-2r) = -\frac{1}{2r+1} \sum_{j=0}^r \binom{2r+1}{2j} B_{2j}(1/2) \zeta(q-2r-1+2j). \quad (4.17)$$

For fixed odd $q \geq 3$, all these moments belong to the same space $\text{span}_{\mathbb{Q}}\{1, \zeta(2), \zeta(4), \dots, \zeta(q-1)\}$. At $q = 1$, formula (4.17) reduces to (4.15).

Proof. For $q \geq 3$ and $x = n + 1/2$, use $H_n^{(q)} = \zeta(q) - \zeta(q, x + 1/2)$ and the midpoint expansion

$$\zeta(q, x + 1/2) = \frac{x^{1-q}}{q-1} + \sum_{j=1}^J \frac{B_{2j}(1/2)(q)_{2j-1}}{(2j)!} x^{1-q-2j} + O(x^{-q-2J-1}).$$

When q is odd, every displayed inverse power is even. Together with the even inverse-power expansion of H_n following its logarithm, this shows that any stated polynomial has an asymptotic expansion containing only $x^{-2j}(\log x)^h$. Finite subtraction in its Dirichlet series leaves possible poles only at $s = 1 - 2j$. This proves regularity at every nonpositive even integer.

To prove the value formula, first take $\Re q > 2r + 2$. The identity $U_q(s) - \zeta(q)\zeta(s, 1/2) = -\sum_{n \geq 0} \zeta(q, n+1)(n+1/2)^{-s}$ continues the difference to $s = -2r$ by an absolutely convergent tail sum. Since $\zeta(-2r, 1/2) = 0$, ordinary summation reversal gives

$$U_q(-2r) = -\sum_{k \geq 1} k^{-q} \sum_{n=0}^{k-1} (n+1/2)^{2r}.$$

The centered Faulhaber identity is

$$\sum_{n=0}^{k-1} (n+1/2)^{2r} = \frac{B_{2r+1}(k+1/2)}{2r+1} = \frac{1}{2r+1} \sum_{j=0}^r \binom{2r+1}{2j} B_{2j}(1/2) k^{2r+1-2j}.$$

Insertion proves (4.17) in this half-plane.

The same tail expansion supplies joint meromorphic continuation in (s, q) ; its possible moving divisors are $s + q = 2 - 2j$. None passes through $(-2r, q)$ for an odd positive integer q . The apparent pole at $q = 1$ cancels: near $s = -2r$ the combination

$$\zeta(q)\zeta(s, 1/2) - \frac{\zeta(s+q-1, 1/2)}{q-1}$$

is jointly holomorphic after writing it as $(\zeta(q) - (q-1)^{-1})\zeta(s, 1/2) - (\zeta(s+q-1, 1/2) - \zeta(s, 1/2))/(q-1)$. The remaining finite subtractions and the normal remainder have no divisor there. Thus meromorphic continuation of the value formula really gives the separately specified odd-order functions, including $H_n^{(1)} = H_n$. Finally, every zeta argument in (4.17) is even; negative even arguments give zero and $\zeta(0) = -1/2$, proving the claimed fixed span. $\square$

For example,

$$\begin{aligned} U_3(0) &= -\zeta(2), & U_3(-2) &= \frac{1}{6} + \frac{1}{12}\zeta(2), & U_3(-4) &= -\frac{1}{12} - \frac{7}{240}\zeta(2), \\ U_5(0) &= -\zeta(4), & U_5(-2) &= -\frac{1}{3}\zeta(2) + \frac{1}{12}\zeta(4), \\ U_5(-4) &= \frac{1}{10} + \frac{1}{6}\zeta(2) - \frac{7}{240}\zeta(4). \end{aligned} \tag{4.18}$$

The odd-order restriction should be retained. At an even-order specialization a vanishing coefficient can still meet a moving divisor, so continuing the fixed spectral value in q need not agree with specializing q before taking the spectral value. No such ambiguity occurs at the odd orders proved here.

4.6 Compact generators at zero, minus two, and minus four

Put $Q(u) = V_0(u)$, $V(u) = V_1(u)$, and $W(u) = V_2(u)$. The first three moment rows have the following compact forms.

Corollary 4.7 (Three complete rows in the harmonic power). *For $|u| < 1$,*

$$(u+1)Q(u) = \frac{u}{2} + \frac{1}{2} \sum_{k \geq 2} (k-1) \frac{u^{k+1}}{(k+1)!} E_k(u), \tag{4.19}$$

$$\begin{aligned} (u+3)V(u) &= \frac{u^3 - 3u - 3}{12} Q(u) - \frac{u^3}{24} \\ &\quad - \frac{1}{24} \sum_{k \geq 2} (k-1)(k+1)(k+6) \frac{u^{k+3}}{(k+3)!} E_k(u), \end{aligned} \tag{4.20}$$

$$\begin{aligned} (u+5)W(u) &= -\frac{u^3 + 18u^2 + 90u + 150}{60} V(u) \\ &\quad - \frac{u^3 + 6u^2 + 15u + 15}{240} Q(u) \\ &\quad + \frac{1}{120} \sum_{k \geq 2} k(k-1)(k+1) \frac{u^{k+5}}{(k+5)!} E_k(u). \end{aligned} \tag{4.21}$$

Proof. The $m = 0$ case of (4.9), followed by substitution of (4.7), is (4.19). For the next row use

$$V = G_2 - G_1 + \frac{1}{4}G_0.$$

The G_1 coefficient is $-u(u+2)/2$ before its substitution, so its apparent denominator $u+2$ cancels. Substituting B gives the factor

$$\frac{k+1}{2(k+3)!} - \frac{1}{24k!} = -\frac{(k-1)(k+1)(k+6)}{24(k+3)!},$$

which proves (4.20). For the last row substitute $m \leq 4$ into

$$W = G_4 - 2G_3 + \frac{3}{2}G_2 - \frac{1}{2}G_1 + \frac{1}{16}G_0.$$

Collect Q , V , the constant part of B , and the coefficient of each E_k separately. The first three are respectively $-(u^3 + 6u^2 + 15u + 15)/(240(u + 5))$, $-(u^3 + 18u^2 + 90u + 150)/(60(u + 5))$, and zero; the last is $k(k - 1)(k + 1)u^{k+5}/(120(u + 5)(k + 5)!)$. These rational simplifications prove (4.21). The verification script replays them with symbolic u and k . $\square$

The zero row (4.19) recovers the recurrence already proved in [5]. Equations (4.20) and (4.21) determine two further complete rows, while Theorem 4.2 supplies every remaining row. For an explicit finite coefficient version, write $Q_p = F_p(0, 1/2)$, $V_p = F_p(-2, 1/2)$ and $Q_0 = V_0 = 0$. For $p \geq 1$, (4.20) says

$$3V_p + pV_{p-1} = \frac{p(p-1)(p-2)}{12}Q_{p-3} - \frac{p}{4}Q_{p-1} - \frac{1}{4}Q_p - \frac{1}{4}\mathbf{1}_{p=3} - \frac{1}{24} \sum_{k=2}^{p-3} (k-1)(k+1)(k+6) \binom{p}{k+3} E_{p-k-3,k}. \quad (4.22)$$

The first term is omitted for $p < 3$, and an empty sum is zero. There is no infinite coefficient calculation at a fixed p .

For example, the next fifth-power identity is

$$F_5(-2, 1/2) = -\frac{200}{81} - \frac{23}{54}\zeta(2) + \frac{5}{12}\zeta(3) - \frac{11}{8}\zeta(4). \quad (4.23)$$

At the next spectral point the first five powers are

$$\begin{aligned} F_1(-4, 1/2) &= \frac{7}{480}, & F_2(-4, 1/2) &= \frac{19}{3600}, \\ F_3(-4, 1/2) &= \frac{493}{18000} + \frac{7}{480}\zeta(2), \\ F_4(-4, 1/2) &= -\frac{6479}{67500} + \frac{19}{1800}\zeta(2) + \frac{7}{80}\zeta(3), \\ F_5(-4, 1/2) &= \frac{98437}{202500} + \frac{643}{5400}\zeta(2) + \frac{19}{240}\zeta(3) + \frac{77}{160}\zeta(4). \end{aligned} \quad (4.24)$$

Coefficient extraction from (4.19)–(4.21) proves these evaluations. Beyond $E_{0,k} = \zeta(k)$, the only low-weight reductions needed are the classical identities $E_{1,2} = \zeta(3)$, $E_{1,3} = \zeta(4)/4$ and $E_{2,2} = 11\zeta(4)/4$, also used in [5].

4.7 Exact exponent singularities and reproducible checks

The apparent negative even denominators in the triangular recursion cancel in every centered row. This has a stronger analytic explanation.

Corollary 4.8 (Exact negative poles of the exponent generator). *On $\Re u < 1$, the function V_r has exactly the simple poles*

$$u = -2r - 1, -2r + 1, \dots, -1.$$

At $u_j = 2j - 2r - 1$, $0 \leq j \leq r$, its residue is

$$\text{Res}_{u=u_j} V_r(u) = -e^{\gamma u_j} c_j(u_j). \quad (4.25)$$

Its Taylor series at zero therefore has exact radius 1.

Proof. The finite subtraction in (4.5) leaves precisely these possible poles in the indicated half-plane. The Hurwitz pole is $\zeta(1 - (u - u_j), N + 1/2)$, with residue -1 in u , proving (4.25). To check that none is removable, use $B_{2j}(1/2) = (2^{1-2j} - 1)B_{2j}$. Thus $b_j = (-1)^{j+1}|b_j|$. For a negative real u , every term in the coefficient $c_j(u)$ has sign $(-1)^j$, and at least one term is nonzero; also $c_0 = 1$. Hence every displayed residue is nonzero. The pole at -1 fixes the exact Taylor radius. $\square$

The accompanying `code/verify_harmonic_powers.py` performs exact rational checks of the three generator rows, extracts the displayed values, checks the first two Bernoulli formulas and the centered Faulhaber identity through multiple moment orders, and evaluates six odd generalized-harmonic checkpoints. Its independent numerical calculation starts from the original harmonic-power sums, subtracts their centered Euler–Maclaurin expansion, and evaluates the remaining Hurwitz order derivatives. Two different truncation lengths are compared. This diagnostic tests signs and constants without reusing the triangular generator; it is not a proof of an infinite identity or a certified numerical error bound. The proof is the normally convergent telescoping argument and the joint continuation established above. The first regular spectral coefficient beyond these centered values still carries global Mellin information; the theorem makes no unsupported claim to eliminate it.

5 An exact geometric collision formula and harmonic hierarchy laws

The incoming mixed-spectral report already determines the collision of three undifferentiated Stieltjes factors at shifts $(0, a\epsilon, b\epsilon)$ for fixed $0 < a < b$. It also supplies the complete spectral subtraction at a fixed, arbitrarily multiple collision [2]. These are substantial antecedents. A moving configuration such as $(0, \epsilon, \epsilon + \epsilon^2)$ is a different question: inserting ϵ -dependent rates in a fixed-rate error estimate is not justified without uniform control. This section proves an exact formula before any rate specialization. It permits arbitrarily many polygamma factors, arbitrary argument-derivative orders, and analytic collision arcs with several different vanishing orders. Its correction is a finite expression in polygamma values, derivatives, rational functions, and logarithms. The corrected remainder is holomorphic in all the shifts.

5.1 The coordinate convention and the local telescoping identity

Write

$$f_r(z) = -\psi^{(r)}(z), \quad n_i = r_i + 1, \quad \kappa_i = (-1)^{r_i} r_i!, \quad h_i(z) = f_{r_i}(1 + z).$$

Thus $f_0 = \gamma_0$ and $f_r = \gamma_0^{(r)}$; this section concerns Stieltjes index zero with arbitrary argument derivatives. In particular, no claim about all positive Stieltjes indices is hidden in the notation. The polygamma recurrence [11] gives the exact local formula

$$f_{r_i}(\{x\}) = h_i(x) + \kappa_i \mathbf{1}_{x>0} x^{-n_i}, \quad 0 < |x| < \delta < 1. \quad (5.1)$$

The functions h_i are holomorphic for $|x| < 1$.

For real shifts $a_1 < \dots < a_d$ sufficiently close to zero, define

$$C_r(\mathbf{a}) = \text{FP}_x \int_0^1 \prod_{i=1}^d f_{r_i}(\{x + a_i\}) \, dx, \quad Q_r = \text{FP}_x \int_0^1 \prod_{i=1}^d f_{r_i}(x) \, dx. \quad (5.2)$$

At the seam $x = -a_i$ on the circle, the cutoff coordinate is exactly $x + a_i$, taken positive, and the deleted right interval has ambient length η_i . The finite part is the coefficient independent of all η_i and $\log \eta_i$. The cutoffs may be taken independently. These are pointwise products away from finitely many distinct seams; there is no multiplication of distributions with overlapping singular supports. The merged quantity uses the same ambient coordinate x .

Put, for $1 \leq m \leq d$,

$$K_m(z, \mathbf{a}) = \frac{\kappa_m}{(z + a_m)^{n_m}} \prod_{k < m} h_k(z + a_k) \prod_{k > m} f_{r_k}(z + a_k). \quad (5.3)$$

In a fixed neighborhood of the merged seam one has

$$\prod_{i=1}^d f_{r_i}(\{x + a_i\}) = \prod_{i=1}^d h_i(x + a_i) + \sum_{m=1}^d \mathbf{1}_{x > -a_m} K_m(x, \mathbf{a}). \quad (5.4)$$

Indeed, expand the product according to the first factor for which the singular part in (5.1) is selected. If this first index is m and $x > -a_m$, then $x > -a_k$ for every $k > m$, so all the subsequent factors may be replaced by their meromorphic germs $f_{r_k}(x + a_k)$. This telescoping form replaces a sum over 2^d subsets by d terms.

For $m < i$ and $1 \leq p \leq n_i$, let $B_{mi,p}(\mathbf{a})$ be the coefficient of $(z + a_i)^{-p}$ in the principal part of K_m at $-a_i$. It is explicitly

$$B_{mi,p}(\mathbf{a}) = \frac{\kappa_i}{(n_i - p)!} \partial_z^{n_i - p} \left\{ \frac{\kappa_m}{(z + a_m)^{n_m}} \prod_{k < m} h_k(z + a_k) \prod_{\substack{k > m \\ k \neq i}} f_{r_k}(z + a_k) \right\} \Big|_{z = -a_i}. \quad (5.5)$$

Only finitely many derivatives are required. All arguments after this substitution are differences of shifts.

5.2 Exact subtraction, with a jointly holomorphic remainder

Theorem 5.1 (Geometric collision formula). *With the convention of (5.2), set*

$$\mathcal{A}_r(\mathbf{a}) = \sum_{m < i} \left\{ -B_{mi,1}(\mathbf{a}) \log(a_i - a_m) + \sum_{p=2}^{n_i} \frac{B_{mi,p}(\mathbf{a})}{(p-1)(a_i - a_m)^{p-1}} \right\}. \quad (5.6)$$

There is a function $\mathcal{R}_r$ holomorphic in all a_i on a complex neighborhood of $\mathbf{a} = 0$ such that, in the stated real ordering chamber,

$$\boxed{\mathcal{C}_r(\mathbf{a}) = \mathcal{A}_r(\mathbf{a}) + \mathcal{R}_r(\mathbf{a}), \quad \mathcal{R}_r(\mathbf{0}) = Q_r.} \quad (5.7)$$

The theorem applies to every number of factors and every tuple of nonnegative derivative orders. It supplies the complete collision correction, including its finite terms, rather than just its leading divergences.

Proof. Choose a fixed small $\delta > 0$ and represent the circle near its seam by $x \in (-\delta, \delta)$. The integral outside this neighborhood and the integral of the all- h_i term in (5.4) are holomorphic in the shifts. Consider one K_m term, integrated from $-a_m$ to δ with its right endpoint cutoff at $-a_m$.

It is important not to take the limits of its individual principal-part coefficients, which can diverge when poles coalesce. Define

$$D_m(z, \mathbf{a}) = \prod_{i=m}^d (z + a_i)^{n_i}, \quad H_m(z, \mathbf{a}) = D_m(z, \mathbf{a}) K_m(z, \mathbf{a}).$$

The numerator is jointly holomorphic: explicitly it is

$$H_m = \kappa_m \prod_{k < m} h_k(z + a_k) \prod_{k > m} \{\kappa_k + (z + a_k)^{n_k} h_k(z + a_k)\}.$$

For distinct shifts, write the polar decomposition

$$K_m(z, \mathbf{a}) = \sum_{i \geq m} \sum_{p=1}^{n_i} \frac{B_{mi,p}(\mathbf{a})}{(z + a_i)^p} + W_m(z, \mathbf{a}),$$

where the notation $B_{mm,p}$ is used here only for the analogous principal coefficients at the starting point. The remainder extends jointly holomorphically as all the shifts merge. For example, on a fixed circle $|w| = \rho > \delta$ inside the domain of all H_m , with the shifts sufficiently small, the residue theorem gives

$$W_m(z, \mathbf{a}) = \frac{1}{2\pi i} \int_{|w|=\rho} \frac{H_m(w, \mathbf{a})}{D_m(w, \mathbf{a})(w-z)} dw, \quad |z| < \rho. \quad (5.8)$$

The denominator stays away from zero on this contour, proving the claimed joint holomorphy. This is the analytic cancellation usually encoded by confluent Hermite divided differences.

A primitive of the polar part is

$$U_m(z, \mathbf{a}) = \sum_{i \geq m} B_{mi,1} \log(z + a_i) - \sum_{i \geq m} \sum_{p=2}^{n_i} \frac{B_{mi,p}}{p-1} (z + a_i)^{1-p}.$$

At the upper point $z = \delta$ its apparent collision poles cancel. To see this without a coefficientwise limit, choose a second, smaller circle $|w| = \rho_0 < \delta$ enclosing all $-a_i$. Then

$$U_m(\delta, \mathbf{a}) = \frac{1}{2\pi i} \int_{|w|=\rho_0} K_m(w, \mathbf{a}) \log(\delta - w) dw. \quad (5.9)$$

The logarithm is holomorphic inside this smaller circle. The displayed integral is jointly holomorphic in the shifts. For $p \geq 2$, the derivative of the logarithm in the residue calculation is $-(p-2)!/(\delta + a_i)^{p-1}$, giving exactly the sign in U_m .

At the lower point $z = -a_m + \eta$, the terms belonging to $i = m$ contain only $\log \eta$ and negative powers of η ; their cutoff constant is zero. For $i > m$, ordinary substitution at $\eta = 0$ is legitimate. Subtracting these lower primitive values gives precisely the m th summand in (5.6). Consequently the remainder of the local integral is

$$U_m(\delta, \mathbf{a}) + \int_{-a_m}^{\delta} W_m(z, \mathbf{a}) dz,$$

which is jointly holomorphic, by (5.8) and (5.9). At zero shifts it is exactly the ambient finite part of $K_m(z, \mathbf{0})$ on $(0, \delta)$: all merged lower polar primitive terms again have zero cutoff constant. Finally sum over m and add the holomorphic all-regular and exterior integrals. The zero-shift form of (5.4) identifies their total with Q_r . This proves (5.7). $\square$

Corollary 5.2 (Analytic collision arcs). *Let $a_i(\epsilon)$ be real analytic, vanish at zero, and be pairwise distinct for sufficiently small $\epsilon > 0$, with a fixed ordering there. Then the complete expansion of $\mathcal{C}_r(\mathbf{a}(\epsilon))$ has the form*

$$U(\epsilon) + V(\epsilon) \log \epsilon,$$

where U and V are convergent Laurent germs with finite principal parts. Its coefficient independent of ϵ and $\log \epsilon$ is

$$\text{CT}_\epsilon \mathcal{C}_r(\mathbf{a}(\epsilon)) = Q_r + \text{CT}_\epsilon \mathcal{A}_r(\mathbf{a}(\epsilon)). \quad (5.10)$$

Every term of the second constant is explicitly determined by finitely many jets of the arcs and of the polygamma functions at 1.

Proof. Each positive gap is $a_i - a_m = b_{im} \epsilon^{\nu_{im}} (1 + O(\epsilon))$, with $b_{im} > 0$ and positive integer ν_{im} . Its logarithm is $\nu_{im} \log \epsilon + \log b_{im}$ plus a convergent Taylor series. Formula (5.5) is meromorphic after composition with these arcs: all possible poles come from a finite number of powers of nonzero analytic gaps. Thus (5.6) has the stated form and its constant needs only finitely many coefficients. The holomorphic remainder contributes exactly its value at zero. No higher powers of $\log \epsilon$ arise in this polygamma case. $\square$

5.3 A short formula at Stieltjes index zero

Let $f = f_0 = -\psi$ and $h(z) = f(1+z)$. If every $r_i = 0$, all poles in (5.3) are simple and (5.6) collapses to

$$\mathcal{A}_0(\mathbf{a}) = \sum_{m < i} \frac{\log(a_i - a_m)}{a_i - a_m} \prod_{k < m} h(a_k - a_i) \prod_{\substack{k > m \\ k \neq i}} f(a_k - a_i). \quad (5.11)$$

The negative arguments here are values of the meromorphic digamma germ, not fractional parts. They are defined because the small gaps are nonzero. Formula (5.11) has only $d(d-1)/2$ summands.

For three factors at shifts $(0, a, b)$ it reads

$$\mathcal{A}_3(0, a, b) = \frac{\log a}{a} f(b-a) + \frac{\log b}{b} f(a-b) + \frac{\log(b-a)}{b-a} h(-b). \quad (5.12)$$

This exact identity is the key to the hierarchical limits below. For fixed $a = \alpha\epsilon$, $b = \beta\epsilon$, expanding

$$h(z) = \gamma + \sum_{k \geq 1} (-1)^k \zeta(k+1) z^k \quad (5.13)$$

recovers every displayed term through order ϵ^0 in the incoming fixed-rate triple theorem [2]. In particular, for $(0, \epsilon, 2\epsilon)$ the finite discrepancy is $\zeta(2) \log 2/2$, as already established there. The addition in (5.12) is the exact correction at arbitrary small a, b , with a jointly holomorphic remainder, so rates may now coalesce.

Even a binary collision illustrates why the parameter matters. For $d = 2$, (5.11) is $\log(a_2 - a_1)/(a_2 - a_1)$. Thus

$$\text{CT}_\epsilon \mathcal{C}_{0,0,0}(0, \epsilon + b\epsilon^2) = Q_{0,0} + b, \quad (5.14)$$

because $\log(\epsilon + b\epsilon^2)/(\epsilon + b\epsilon^2) = \epsilon^{-1} \log \epsilon - b \log \epsilon + b + O(\epsilon |\log \epsilon|)$. This does not change the preceding binary theorem in its stated shift coordinate. It records the effect of choosing a nonlinear collision arc.

5.4 Two different hierarchies and their harmonic constants

Set $Q_3 = Q_{0,0,0}$, and write $H_n = \sum_{j=1}^n 1/j$ for $n \geq 1$. In the formulas below a term containing $H_{u/v}$ is included only when v divides u ; no value at a nonintegral index is needed.

Theorem 5.3 (Oriented harmonic hierarchy laws). *Let $p > q \geq 1$ be integers and $v = p - q$. The left nested collision obeys*

$$\text{CT}_\epsilon \mathcal{C}_{0,0,0}(0, \epsilon^p, \epsilon^q) = Q_3 - \mathbf{1}_{v|q} \gamma H_{q/v}. \quad (5.15)$$

The right nested collision has the different value

$$\begin{aligned} \text{CT}_\epsilon \mathcal{C}_{0,0,0}(0, \epsilon^q, \epsilon^q + \epsilon^p) = & Q_3 + \mathbf{1}_{v|p+q} (-1)^{(p+q)/v} H_{(p+q)/v} \\ & + \mathbf{1}_{v|q} \gamma (-1)^{q/v+1} H_{q/v}. \end{aligned} \quad (5.16)$$

In particular,

$$\text{CT}_\epsilon \mathcal{C}_{0,0,0}(0, \epsilon^2, \epsilon) = Q_3 - \gamma, \quad (5.17)$$

$$\text{CT}_\epsilon \mathcal{C}_{0,0,0}(0, \epsilon, \epsilon + \epsilon^2) = Q_3 + \gamma - \frac{11}{6}, \quad (5.18)$$

$$\text{CT}_\epsilon \mathcal{C}_{0,0,0}(0, \epsilon, \epsilon + \epsilon^3) = Q_3 + \frac{3}{2}. \quad (5.19)$$

Proof. The two elementary, convergent generating functions needed are

$$\frac{\log(1-z)}{1-z} = -\sum_{n \geq 1} H_n z^n, \quad \frac{\log(1+z)}{1+z} = \sum_{n \geq 1} (-1)^{n+1} H_n z^n. \quad (5.20)$$

The first follows by multiplying $-\log(1-z) = \sum_{j \geq 1} z^j/j$ by the geometric series; the second is obtained by replacing z with $-z$.

For the left nested configuration, substitute $a = \epsilon^p$, $b = \epsilon^q$ into (5.12). The first two logarithms are respectively $p \log \epsilon$ and $q \log \epsilon$. They therefore make no contribution to the log-free constant, regardless of the poles of their other factors. The only possible contribution is

$$\epsilon^{-q} \frac{\log(1-\epsilon^v)}{1-\epsilon^v} h(-\epsilon^q) = -\sum_{n \geq 1} H_n \epsilon^{nv-q} \left\{ \gamma + \sum_{k \geq 1} \zeta(k+1) \epsilon^{qk} \right\}.$$

For $k \geq 1$, the exponent $nv - q + qk$ is strictly positive. The constant comes solely from $k = 0$ and $nv = q$, proving (5.15) after (5.10) is used.

For the right nested configuration, put $a = \epsilon^q$, $b = \epsilon^q + \epsilon^p$. Now the first and third logarithms in (5.12) are pure multiples of $\log \epsilon$. The log-free part of the second term is

$$\begin{aligned} & \frac{\log(1+\epsilon^v)}{1+\epsilon^v} \epsilon^{-q} f(-\epsilon^p) \\ &= \left\{ \sum_{n \geq 1} (-1)^{n+1} H_n \epsilon^{nv} \right\} \left\{ -\epsilon^{-p-q} + \gamma \epsilon^{-q} + \sum_{k \geq 1} \zeta(k+1) \epsilon^{pk-q} \right\}. \end{aligned}$$

All terms in the last sum have positive exponents. The pole term contributes only if $nv = p + q$, with coefficient $(-1)^n H_n$; the constant regular term contributes only if $nv = q$, with coefficient $\gamma(-1)^{n+1} H_n$. This proves (5.16). The examples use $H_1 = 1$, $H_2 = 3/2$, and $H_3 = 11/6$. $\square$

These two arrangements are not interchangeable. The original periodic Stieltjes germ has a one-sided singularity, and reflection is not a symmetry of its regular part. For example, subtracting (5.17) from (5.18) eliminates the unevaluated merged moment and gives the exact difference $2\gamma - 11/6$. The fixed-rate error estimate in the preceding work alone would not establish either formula.

5.5 A quartic example and an evaluated polygamma constant

For linear rates $a_i = c_i \epsilon$, $c_1 < \dots < c_d$, define the finite Laurent coefficient

$$\alpha_{mi}(\mathbf{c}) = \frac{1}{c_i - c_m} [\epsilon^1] \left\{ \prod_{k < m} h((c_k - c_i)\epsilon) \prod_{\substack{k > m \\ k \neq i}} f((c_k - c_i)\epsilon) \right\}. \quad (5.21)$$

Then the constant-order part of the complete counterterm is

$$[\epsilon^0] \mathcal{A}_0(\epsilon \mathbf{c}) = \sum_{m < i} \alpha_{mi}(\mathbf{c}) \{ \log \epsilon + \log(c_i - c_m) \}. \quad (5.22)$$

In particular, its log-free constant is the finite sum with $\log(c_i - c_m)$. It requires only (5.13) through degree $d - 2$. For rational rates every coefficient is a rational polynomial in $\gamma, \zeta(2), \dots, \zeta(d - 1)$; its formal weight is $d - 1$ when γ has weight one and $\zeta(k)$ has weight k . Indeed, a coefficient multiplying z^j in either germ has weight $j + 1$, including the pole coefficient at $j = -1$; there are $d - 2$ germ factors and their exponents sum to one in (5.21). This is a statement about the produced expressions, not an assertion of arithmetic independence.

For the quartic rates $(0, 1, 2, 3)$ the six coefficients are

(m, i)	$c_i - c_m$	α_{mi}
(1, 2)	1	$-3\gamma\zeta(2) + \frac{9}{2}\zeta(3)$
(1, 3)	2	0
(1, 4)	3	$\gamma\zeta(2) - \frac{3}{2}\zeta(3)$
(2, 3)	1	$\gamma\zeta(2) + 4\zeta(3)$
(2, 4)	2	$2\gamma\zeta(2) - \frac{9}{2}\zeta(3)$
(3, 4)	1	$5\gamma\zeta(2)$

Consequently, with $Q_4 = Q_{0,0,0,0}$,

$$\begin{aligned} \text{CT}_\epsilon \mathcal{C}_{0,0,0,0}(0, \epsilon, 2\epsilon, 3\epsilon) &= Q_4 + \gamma\zeta(2)(2\log 2 + \log 3) \\ &\quad - \frac{3}{2}\zeta(3)(3\log 2 + \log 3). \end{aligned} \quad (5.23)$$

The coefficient of $\epsilon^0 \log \epsilon$ is $6\gamma\zeta(2) + 5\zeta(3)/2$. A rate difference removes Q_4 :

$$\begin{aligned} \text{CT}_\epsilon \mathcal{C}_{0,0,0,0}(0, \epsilon, 2\epsilon, 4\epsilon) - \text{CT}_\epsilon \mathcal{C}_{0,0,0,0}(0, \epsilon, 2\epsilon, 3\epsilon) \\ = \gamma\zeta(2) \left(\frac{7}{2} \log 2 + \log 3 \right) - \frac{\zeta(3)}{6} (\log 2 + 7 \log 3). \end{aligned} \quad (5.24)$$

Repeated poles add the rational primitive terms in (5.6); retaining only its logarithmic terms would give an incorrect finite constant. For example,

$$\boxed{\text{CT}_\epsilon \text{FP}_x \int_0^1 \gamma_0(x) \gamma'_0(\{x + \epsilon\}) \gamma_0(\{x + 2\epsilon\}) \, dx = 2\gamma\zeta(2) - \zeta(3) \log 2.} \quad (5.25)$$

To check every term, for $\mathbf{r} = (0, 1, 0)$ and rates $(0, 1, 2)$, the three pairs in (5.6) have constant-order contributions

$$(m, i) = (1, 2) : -3\zeta(3) \log \epsilon + \zeta(3), \quad (1, 3) : -\zeta(3)(\log \epsilon + \log 2), \quad (2, 3) : 4\zeta(3) \log \epsilon.$$

The nonlogarithmic $\zeta(3)$ in the first pair comes from its double pole. Hence the finite discrepancy from the merged moment is $\zeta(3)(1 - \log 2)$. The merged value follows directly from the fundamental theorem of calculus before taking the cutoff constant:

$$Q_{0,1,0} = \frac{1}{3} \left\{ \gamma^3 - \text{CT}_{x \downarrow 0} \gamma_0(x)^3 \right\} = 2\gamma\zeta(2) - \zeta(3),$$

since the constant coefficient of $(x^{-1} + \gamma - \zeta(2)x + \zeta(3)x^2 + \dots)^3$ is $\gamma^3 - 6\gamma\zeta(2) + 3\zeta(3)$. Adding the discrepancy proves (5.25) without an unevaluated global moment.

5.6 Verification and scope of the result

The supplied `code/verify\_collisions.py` checks both harmonic hierarchy laws by exact coefficient extraction for every $1 \leq q < p \leq 12$ and verifies the quartic expressions. Its numerical calculation independently integrates the original periodic digamma products, using local Taylor subtraction at each separated seam. The merged moments are obtained from a local Laurent expansion and ordinary quadrature away from zero. The collision formula itself is not used in either quadrature. At 70-digit working precision, selected residuals $\mathcal{C} - \mathcal{A} - Q$ are

Configuration	$\epsilon = 10^{-2}$	$\epsilon = 10^{-3}$
$(0, \epsilon, 2\epsilon, 3\epsilon)$	$-1.468167310 \cdot 10^{-1}$	$-1.501685165 \cdot 10^{-2}$
$(0, \epsilon^2, \epsilon)$	$-1.288859769 \cdot 10^{-2}$	$-1.210723679 \cdot 10^{-3}$
$(0, \epsilon^3, \epsilon^2)$	$-1.202913331 \cdot 10^{-4}$	$-1.202065558 \cdot 10^{-6}$
$(0, \epsilon, \epsilon + \epsilon^2)$	$-1.268176013 \cdot 10^{-2}$	$-1.208676388 \cdot 10^{-3}$

These residuals are consistent with a holomorphic remainder, whose first change is of the order of the largest shift. They are not claimed to be interval certificates. A separate repeated-pole quadrature in `code/verify\_polygamma\_collisions.py` checks (5.25) using second-order local subtraction; the corresponding residuals are approximately 0.02934754 and 0.00224172. Exact coefficient generation for that example and additional derivative placements is supplied in `code/derive\_polygamma\_collisions.py`.

The proof of unrestricted multiplicity, argument-derivative order, and analytic collision rates is the telescoping and contour argument above. The numerical evidence checks its implementation on specified cases. The prior fixed-rate triple discrepancy and fixed-collision spectral completion remain credited to [2]. Positive Stieltjes indices introduce logarithmic singular germs rather than meromorphic ones; a corresponding exact geometric formula would require an additional argument and is not asserted here. The unresolved extension is precisely formulated in the research agenda.

6 Source audit, proposed qualifications, and verification

6.1 A fixed source record

The main source baseline is the ProveIt commit whose full identifier is

921b58f1099568bfefa28a556470e4b29e94eb3b.

It contains the manuscript directory supplied for this task and four incoming archives dated 10–11 October 2026: the bilinear Mellin–Hurwitz, cubic harmonic mixed-order, exact resonant, and independent-order reports [1, 7, 6, 5, 4]. The earlier mixed spectral report, which states several questions answered here, was inspected at the historical revision

5a790187c8e186e41e2b990b4941cb7a1a3c7b6b.

Its archived source is cited separately as [2]; earlier exact identities used in the shift calculation are [3]. This distinction matters because an incoming archive can be integrated and removed from a later incoming directory without invalidating its historical role.

The audit covered the definitions, proofs, status statements, and questions directly used in this continuation, together with the current Gaussian conjecture labels. It did not certify every line of the entire multi-volume manuscript. No new false theorem was identified in the selected source material. The qualifications below state how the new results must be recorded; they are not allegations that the source already contains each invalid inference.

6.2 Status changes justified by the proofs

The shift-boundary question is answered. The mixed spectral report asks for the actual germ at $a = -r$ and the exception at nonpositive spectral orders. Theorems 2.2, 2.4, 2.6, and 2.8 answer it. The existing half-plane theorem remains correct, but a status summary can now give the exact first singularity and exact Taylor radius. Transport itself should be attributed to the broader relative ordered Hurwitz character: the exact identification is (2.5). Treating the present family as unrelated to that source would duplicate an established construction.

The weight-six and weight-eight separation question is answered within its specified family. Equations (3.9) and (3.11) give the requested rational elimination. Theorem 3.2 supplies the general rank, while Theorem 3.4 determines the full summation kernel. The

individual nonlinear coordinate moments at weights six and eight lie outside this formal mixed-Bell span. This does not rule out evaluating them using another parameter family, an argument derivative, or an independent functional identity.

The compact harmonic-power target has a constructive solution. Theorem 4.2 and Corollary 4.7 answer the complete minus-two sequence requested in Question 10 of [5]; they also give every further centered even moment. Theorem 4.4 adds the stronger conclusion that one finite Euler-sum span works for all moment orders at a fixed power. A closed bivariate expression and minimal arithmetic coordinates remain open. Corollary 4.6 supplies a sufficient generalized-harmonic parity class toward Question 11 and a finite formula for every single positive odd harmonic order; it does not finish the requested necessity classification.

Nested collision constants are now explicit in the polygamma sector. The exact formula of Theorem 5.1 supplies a jointly holomorphic remainder for any number of factors and any nonnegative argument-derivative orders. Theorem 5.3 answers the named nested triple paths and all their monomial-rate analogues. The extension to positive Stieltjes indices and a theorem about composition of renormalized merging operations are still separate questions. They should not be marked solved by the present result.

6.3 Qualifications that prevent incorrect extensions

Principal continuation is a statement about a branch. The principal family is holomorphic on the slit plane in (2.15). Earlier removable shifts need not remain removable on every sheet. A loop around $a = -1$ changes D_1 by

$$(e^{-2\pi i s} - 1) \frac{(a+1)^{-s}}{a}.$$

The added term generally has a pole at $a = 0$. A claim of unrestricted removability on the universal continuation would therefore need correction. The source's valid half-plane result makes no such claim.

Polynomial values do not imply polynomial spectral derivatives. The finite Stirling formula at $s = -m$ has rank $m + 1$ in depth. Differentiating the analytic family introduces $z^m(-\log z)^k$ at the first boundary. For every $k \geq 1$ it has finite shift Taylor radius and supplies independent depth functions. It cannot be obtained by differentiating a formula whose label m has only been specified at integers. Similarly, the first spectral derivative of the raw harmonic-power family retains a global Mellin remainder that is absent from its regular value.

A removable restriction need not be a jointly removable germ. The auxiliary G_m in Section 4 obeys $G_m(0) = 0$, whereas separately specializing the uncentered family at exponent zero gives $\zeta(-m)$. The explicit term in Remark 4.3 proves the discrepancy. Its polar hyperplane is not removable merely because its numerator vanishes at one point. In contrast, the centered family has no polar hyperplane through $(-2r, 0)$, so the claimed coefficient extraction is valid.

A collision constant depends on the full primitive. The terms of order $p \geq 2$ in (5.6) are rational primitive terms, in addition to logarithms. They are essential: the derivative example (5.25) would be wrong without them. Even a curved binary gap $\epsilon + b\epsilon^2$ changes the raw constant by b , as (5.14) shows. Normalizing only a leading pole or only a fixed-rate approximation does not determine the required constant along a nested path.

Formal independence and arithmetic independence are distinct. The Bell rank treats the L_k as independent indeterminates, and the shift rank treats analytic functions of a over the constants. The Euler sums in $\mathcal{E}_{p-1}$ form a proved spanning list. None of these assertions gives a lower bound for the arithmetic dimension of their numerical values. The exact matrices are suitable certificates for the specified algebraic problem and no larger one.

The retained Gaussian conjectures remain conjectures. The pinned manuscript labels the current S_6 candidate `cycloquot:conj:S6` in `chapters/04-cyclotomic-quotients.tex`, and the revised S_8 candidate `s8new:conj:S8` in `chapters/04-S8-candidate.tex` [1]. This article proves neither numerical period identity. The rejected older S_8 vector is already distinguished in the current source. No new integer-relation fit or conjectural coefficient vector is offered here. The remaining task is a new exact functional relation with its analytic endpoint evaluation.

6.4 Reproducible exact checks and independent diagnostics

Every primary analytic result has a proof in the preceding sections. The following additional checks are included as readable Python source and JSON results. They require only Python, SymPy, and mpmath. The driver `code/run_all.py` runs the checks in a fixed order.

Family	Recorded verification
Shift germs	85 exact core checks and 50 numerical comparisons: Newton versus defining sums, finite transport, Cauchy-extracted boundary constants, spectral jets, and a doubled cutoff. The primitive supplement adds 12 exact derivative and partial-fraction checks.
Mixed Bell moments	Independent coefficient constructions and ranks for weights 2 through 18; exact elimination and annihilators at weights 6, 8; the quartic zero row; and a deliberately corrupted coefficient that is correctly rejected. Nineteen mixed rows at five real or complex parameter choices are checked numerically, plus the independent harmonic tail-square identity.
Harmonic powers	Seven exact rational generator residuals; 26 Bernoulli identities through moment order $r = 12$; 15 centered values at powers 1–5 and moment orders 0–2; and six full exponent-generator comparisons at two exponent values. The centered computations use two truncation lengths.
Generalized harmonics	Nine exact centered Faulhaber identities and six independent values for harmonic orders $q = 3, 5$ at moment orders $r = 0, 1, 2$, also using two subtraction truncations.
Collisions	132 exact oriented hierarchy assertions for $1 \leq q < p \leq 12$, exact quartic coefficients and rate differences, and the derivative constant. Independent local-subtraction quadrature is used for 12 digamma configurations and two polygamma configurations. The corrected remainder tends to the merged value as predicted.

The largest scaled residual in the core shift comparisons is about $1.76 \cdot 10^{-23}$ and is controlled by the chosen Newton cutoff. The Bell mixed-row comparisons are within $2.70 \cdot 10^{-65}$ in the recorded scaled metric; the tail-square comparison has absolute residual below $8.30 \cdot 10^{-75}$. For the 15 centered harmonic-power values, the largest absolute discrepancy is below $6.49 \cdot 10^{-45}$; the independent truncations differ by at most $1.29 \cdot 10^{-32}$. These are recorded diagnostic values, not rigorous error estimates. The precision, truncation parameters, and individual residuals are retained in the result files.

The collision diagnostics test the exact correction $\mathcal{C} - \mathcal{A}$ against its proved holomorphic limit. They do not assert that this difference equals its limit at a nonzero shift: the remaining Taylor terms are expected. This makes their numerical interpretation different from the residual of an exact finite-parameter Bell identity.

6.5 Integration order

The source sections can be integrated independently, retaining their label prefixes `hsh:`, `bell:`, `hpnew:`, and `hc:`. First reconcile the relative-character identification, then add the shift singularities and primitives; the Bell and harmonic-power sections can follow their existing antecedents; finally insert the collision theorem before the nested-rate examples. The plain-text integration note records exact source dependencies and suggested status replacements. The complete standalone TeX and its compiled PDF are included, alongside the modular source and verification programs.

7 Further research questions and exact targets

The questions below use the new results as starting points. Each asks for a functional identity, a finite reduction, or a specified algebraic classification. None is asserted as a proved conclusion of this article.

7.1 Harmonic powers and generalized harmonic numerators

Question 7.1 (A single generating expression in both orders). The triangular generator determines every $V_r(u)$ using $2r + 1$ steps. Find a useful closed expression for

$$\sum_{r \geq 0} V_r(u) \frac{t^{2r}}{(2r)!}$$

in a stated domain, with its singular terms in t separated and its continuation in u controlled. A satisfactory expression should recover both the negative odd exponent poles and the uniform finite Euler-sum span without first expanding all the triangular rows. The centered Mellin kernel and its even local coefficients give a concrete route. An identity of formal coefficient series should be distinguished from a normally convergent two-variable identity.

Question 7.2 (Minimal coordinates at a fixed harmonic power). For each p , determine the span of all $F_p(-2r, 1/2)$ inside the finite displayed space $\mathcal{E}_{p-1}$. Compute rational coefficient matrices for successive r and reduce them modulo an explicitly stated set of proved multiple-zeta relations. The first targets are $p = 6, 7, 8$, where the generating theorem is finite but individual Euler sums begin to interact nontrivially. The desired output is a minimal presentation relative to those proved relations, not an unsupported assertion of arithmetic independence.

Question 7.3 (A criterion for regular generalized-harmonic products). Classify polynomials in $H_n^{(q)}$ whose centered Dirichlet series are holomorphic at every nonpositive even spectral point. The asymptotic coefficients in inverse powers of $n + 1/2$ give necessary local tests. Determine which combinations involving even harmonic orders cancel all obstructing coefficients, and express the answer by a generating identity rather than an expanding list of examples. This continues Question 11 of [5] and should include both sufficient structural families and a clearly specified necessity statement.

Question 7.4 (Exact cancellation of the first global spectral term). Find finite combinations of the centered harmonic-power families whose first spectral derivatives at $s = -2r$ reduce to a

finite Euler-sum and Stieltjes expression. The regular values alone cannot settle this question: the holomorphic Mellin remainder contributes after differentiation. A proof should first exhibit an exact identity among the Mellin kernels that cancels that remainder, then differentiate the completed family. A first target is a relation among powers $p \leq 4$ at $s = 0$ or $s = -2$.

7.2 Shift singularities, monodromy, and primitives

Question 7.5 (The full relative-character singularity pattern). Extend the explicit scalar amplitudes in Theorem 2.2 from words $(s, \{1\}^r)$ to arbitrary independent complex orders in the relative Hurwitz character of [4]. At each negative shift, determine a finite set of local power-log germs and their exact coefficients. Then identify which words have removable earlier shifts and which exceptional order loci reduce the functional rank. The source transport is available; the challenge is organizing the cancellations without hiding them in a recursively defined remainder.

Question 7.6 (A monodromy representation with explicit generators). The local formula gives the monodromy around each negative integer. Describe the action of finite products of these loops on the span of relative-character specializations, including the rational poles created on nonprincipal sheets. Determine whether a finite-depth filtration is preserved and compute its associated graded action. This would give a global continuation statement that retains the branch information responsible for the exact-radius theorem.

Question 7.7 (Global normalized primitives and shift transport). The local primitive in Section 2.4 is explicit in Lerch functions, with finite polylogarithmic formulas for spectral jets at zero. Construct global primitives normalized at a positive shift, and determine how their constants transform under $a \mapsto a + 1$ and under monodromy. A useful next target is a closed difference equation for the first primitive of D_1 and its first two spectral derivatives, retaining every logarithmic contact term. Repeated primitives may connect to higher Gamma functions, but any such identification must fix the same base-point constants.

7.3 Polygamma moments beyond the two-parameter rows

Question 7.8 (Additional generators that isolate the missing moments). The exact Bell matrices identify the missing formal coordinates at weights six and eight. Find an additional Gamma or beta parameter identity whose mixed derivatives contribute a row outside that span. The immediate target at weight six is an individual value of

$$\sum_{n \geq 0} (n + a/2) L_2(n; a)^3$$

for generic a , since the two combinations in (3.18)–(3.19) then separate the other cubic-weight products. At weight eight, choose candidate generators by pairing their coefficient rows with the explicit annihilators in (3.12) before carrying out numerical experiments. This gives an exact test of whether a proposed family adds information.

Question 7.9 (A hierarchy of generalized-harmonic tail identities). Specialize the full summation kernel in (3.17) at integral and half-integral a . Classify the resulting multiple-zeta or colored multiple-polylogarithm identities at successive weights and compare their span with stated shuffle and stuffle relations. The weight-seven tail-square evaluation is the first compact example. The target is a finite family valid at arbitrary weight, together with a transparent map from Bell rows to convergent harmonic sums, rather than a list of numerically fitted reductions.

7.4 Collision constants and higher Stieltjes indices

Question 7.10 (Positive Stieltjes indices with moving gaps). Extend Theorem 5.1 to products of $\gamma_{m_i}^{(r_i)}(\{x + a_i\})$ with some $m_i > 0$. Their singular pieces contain logarithms, so a meromorphic

partial-fraction decomposition alone is insufficient. Introduce independent local exponents, integrate before differentiating, and seek a jointly holomorphic remainder after subtraction of the full parameter-dependent primitives. The first target is three factors with indices $(1, 0, 0)$ along both nested paths of Theorem 5.3. The statement should specify the allowed logarithm branches and every intermediate normalization.

Question 7.11 (Composition of renormalized collisions). The exact correction isolates a holomorphic remainder at the full collision. Can one define partial-cluster corrections so that merging two clusters and then a third produces the same normalized result as the other parenthesization? Raw constant extraction is already nonassociative in the relevant sense, as the oriented harmonic laws show. Determine the exact defect and whether a local correction removes it. Start with four digamma factors and two nested clusters; the primitive coefficients in (5.5) give finite algebraic data for testing the compatibility equations.

Question 7.12 (Global multiple kernels after local collision subtraction). The present theorem isolates all local gap dependence, leaving the merged constant Q_r . Identify permutation or argument-derivative combinations for which this global constant reduces to endpoint data, extending the evaluated example (5.25). For the remaining combinations, give an explicit finite decomposition into convergent multiple Tornheim-type kernels. Begin with four factors and small total argument-derivative order; distinguish a local completion theorem from an evaluation of the surviving global period.

7.5 The unresolved Gaussian candidates

Question 7.13 (An exact functional bridge for the frozen S_6 and S_8). Retain the current source coefficient vectors and the known formal nonmembership results [1, 5]. Seek an additional iterated-integral, distribution, or functional relation outside the already excluded finite relation families. Any claimed proof should supply both an exact rational certificate and the analytic identity represented by each row, including endpoint regularization. A new relation search inside an unchanged excluded span cannot settle the conjecture. The Bell and harmonic identities here provide exact auxiliary periods, but no bridge from them to the Gaussian targets has been established.

The most immediate extensions are the fixed-power coordinate matrices, the first positive-Stieltjes-index nested collision, and a new Gamma generator selected using the weight-six annihilator. Each begins from a finite, explicit calculation and asks for an exact functional identity. The broader monodromy and Gaussian questions require additional structural input.

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